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Blank et al.

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(54) **AGRICULTURAL HARVESTING MACHINE WITH PRE-EMERGENCE WEED DETECTION AND MITIGATION SYSTEM**

(58) **Field of Classification Search**
None
See application file for complete search history.

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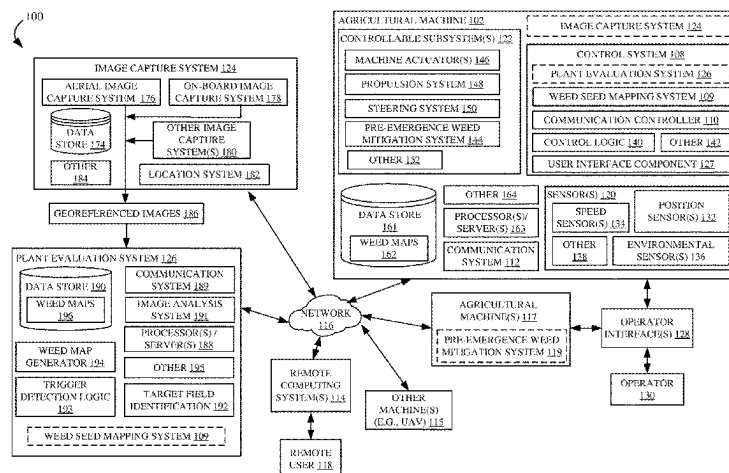
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(57) **ABSTRACT**

An agricultural harvesting machine includes crop processing functionality configured to engage crop in a field, perform a crop processing operation on the crop, and move the processed crop to a harvested crop repository, and a control system configured to identify a weed seed area indicating presence of weed seeds, and generate a control signal associated with a pre-emergence weed seed treatment operation based on the identified weed seed area.

20 Claims, 16 Drawing Sheets



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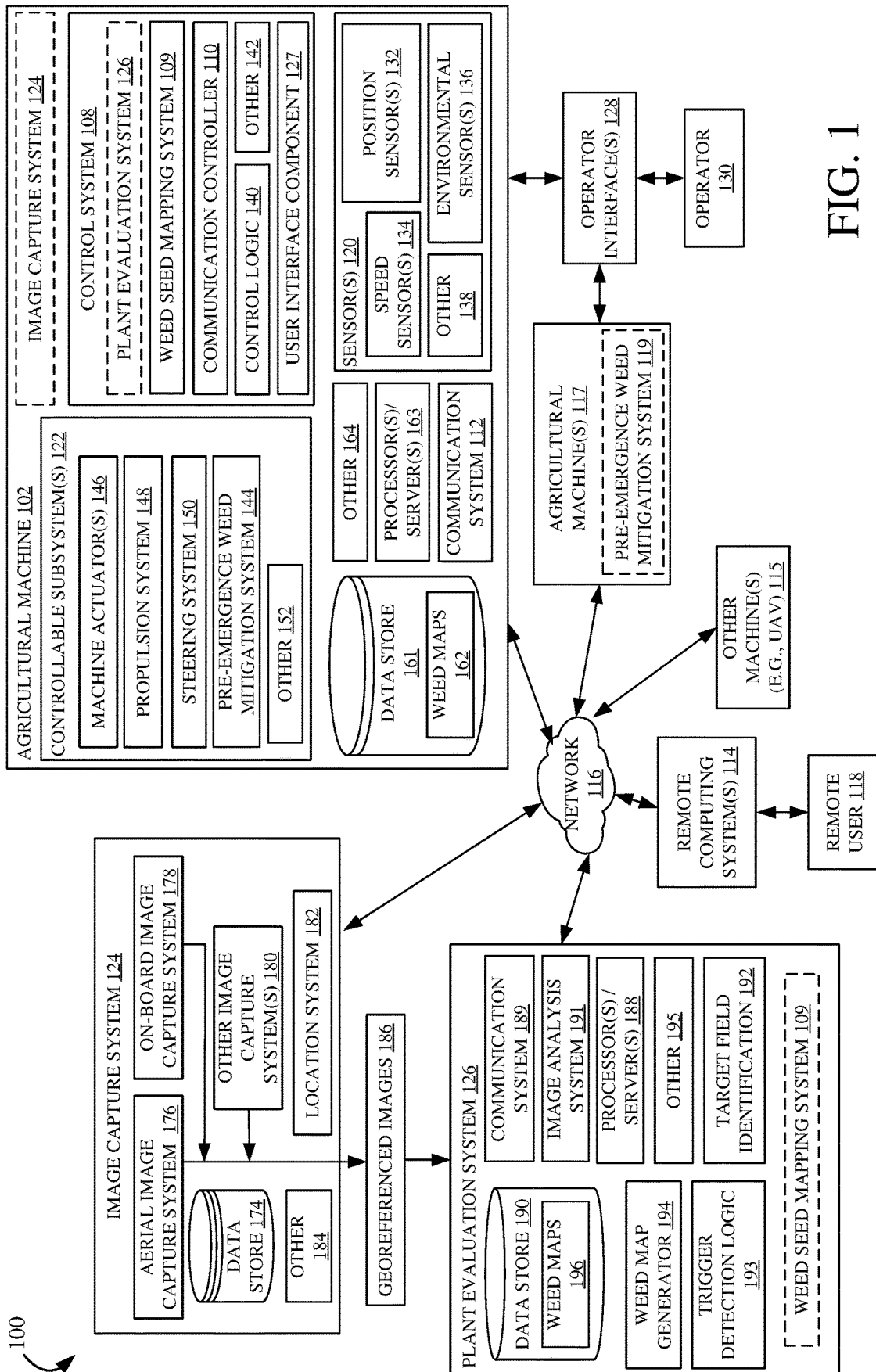


FIG. 1

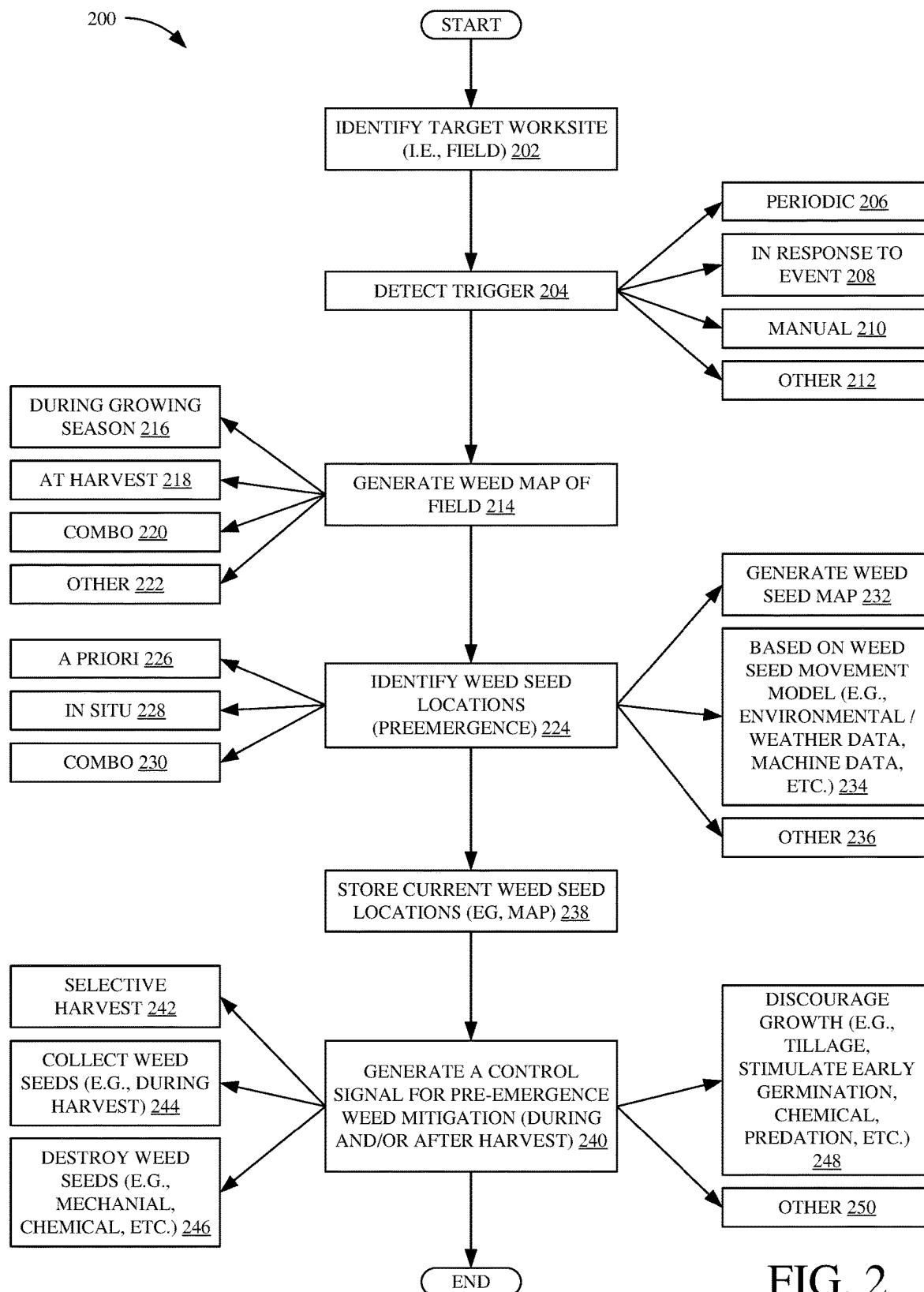


FIG. 2

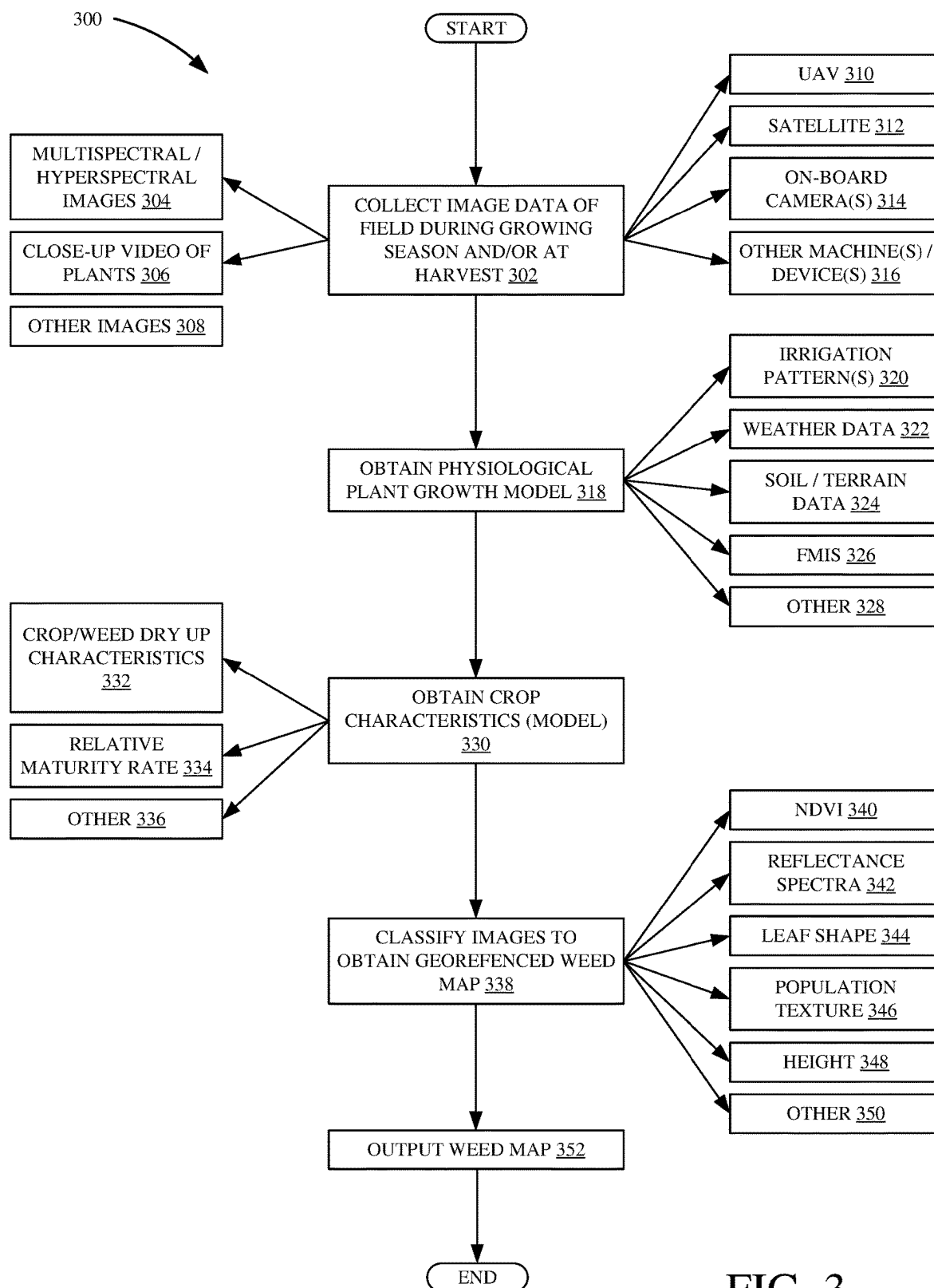


FIG. 3

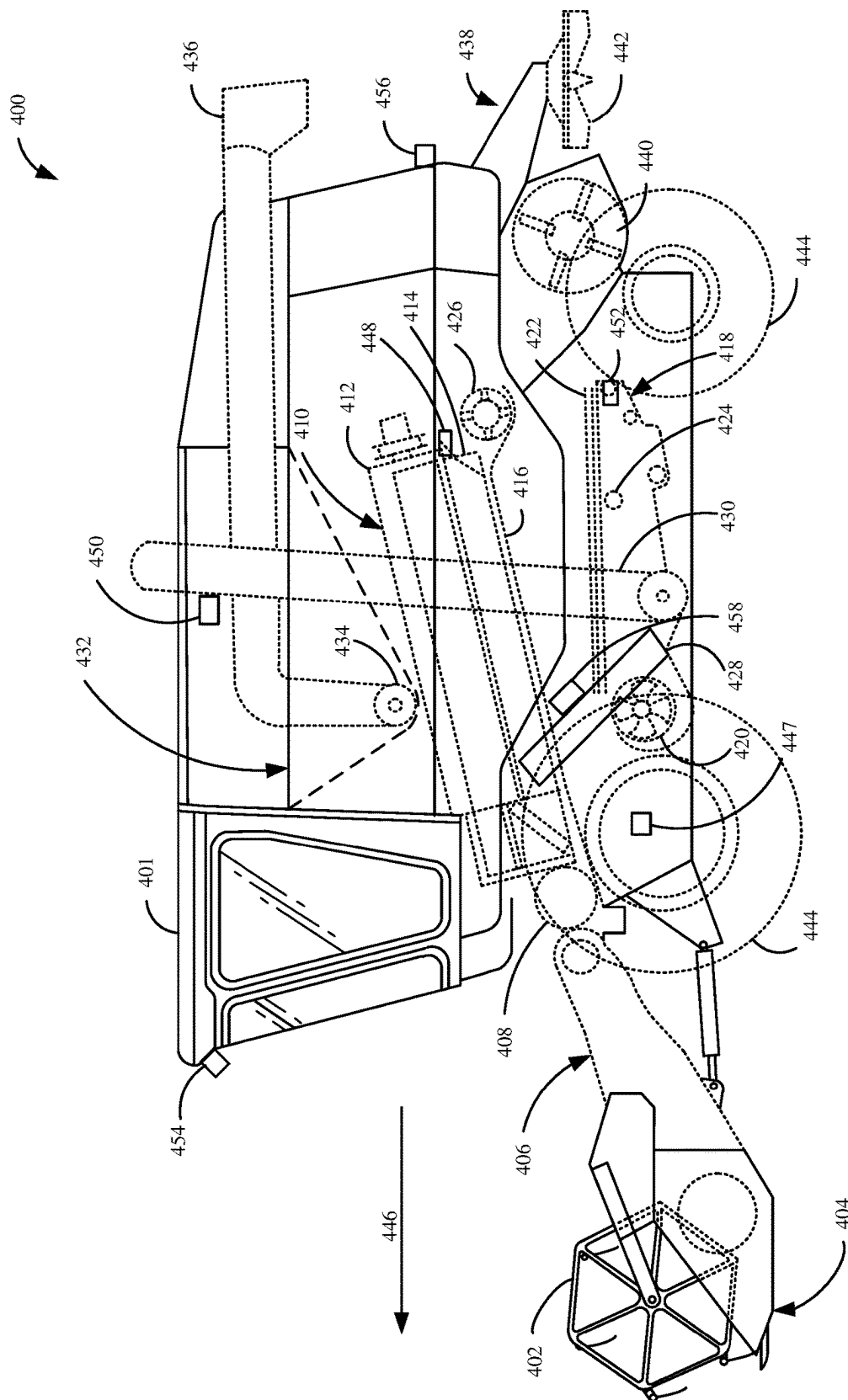


FIG. 4

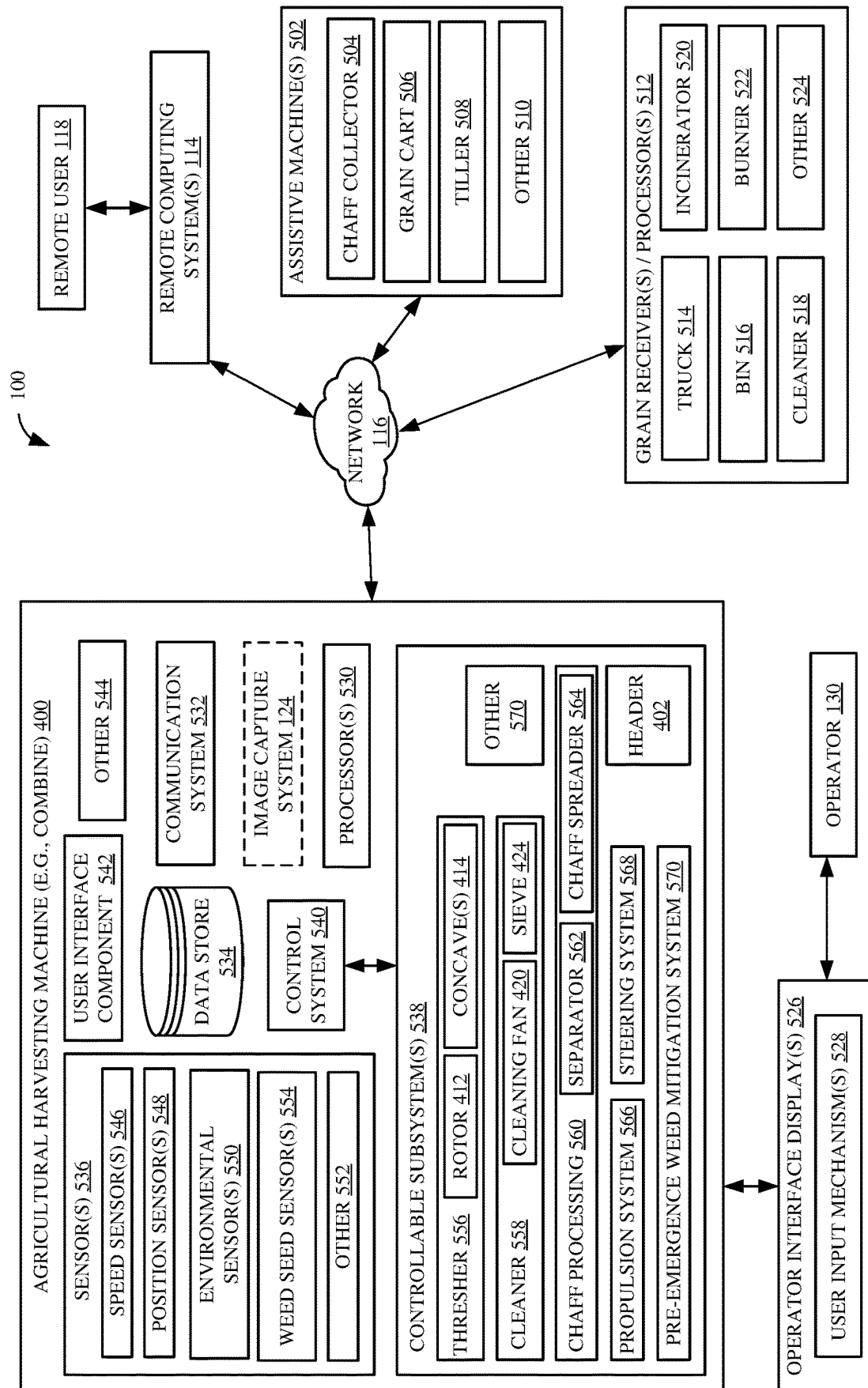


FIG. 5

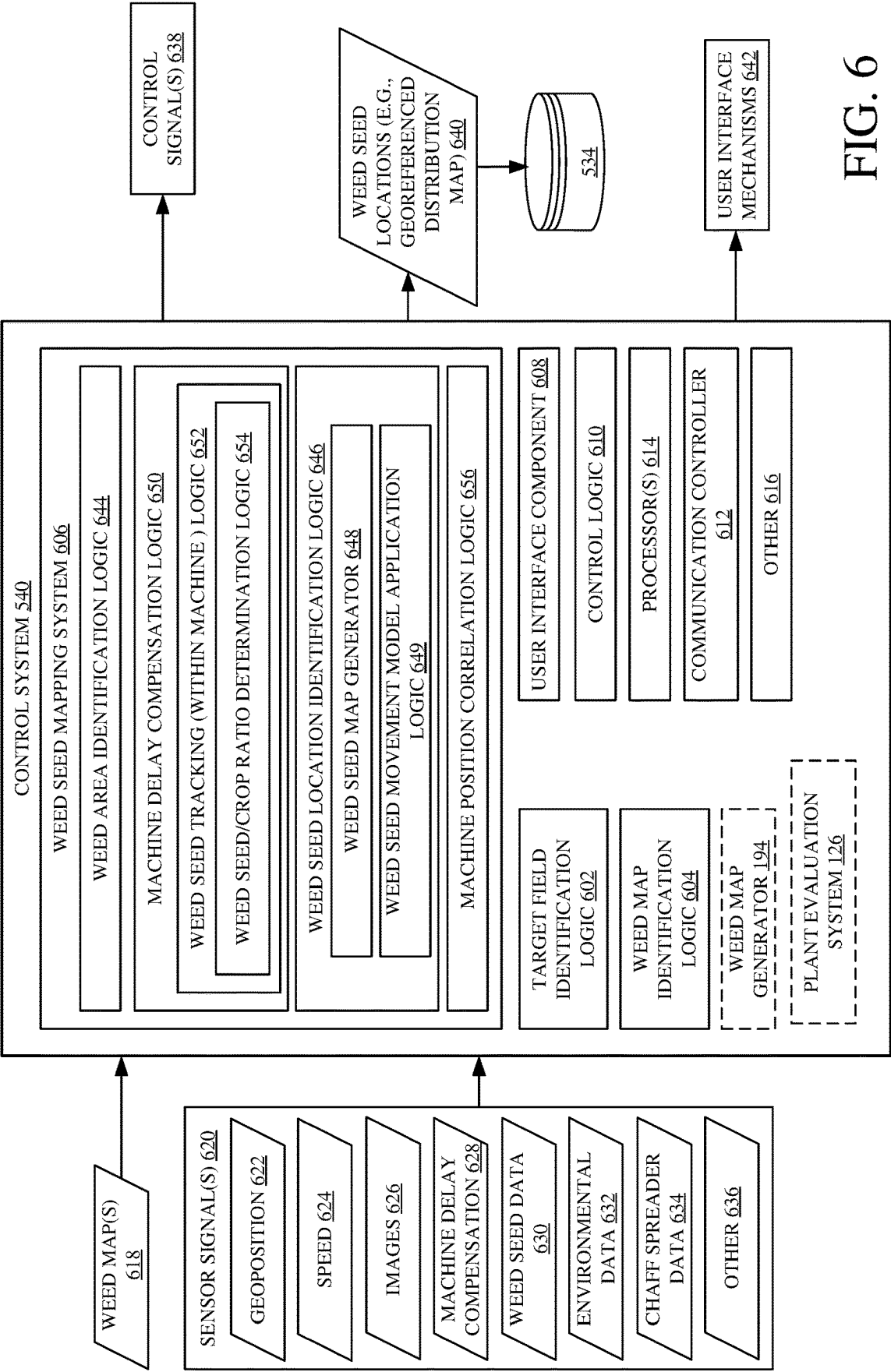


FIG. 6

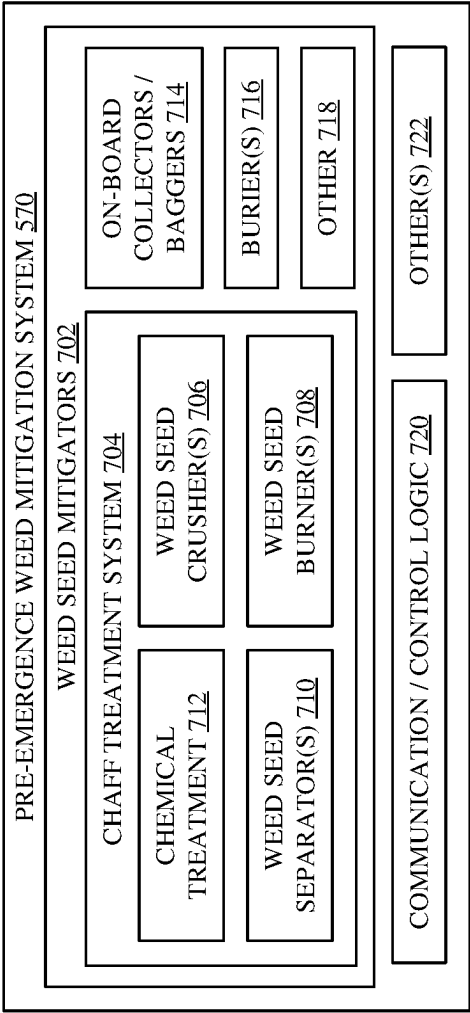


FIG. 7

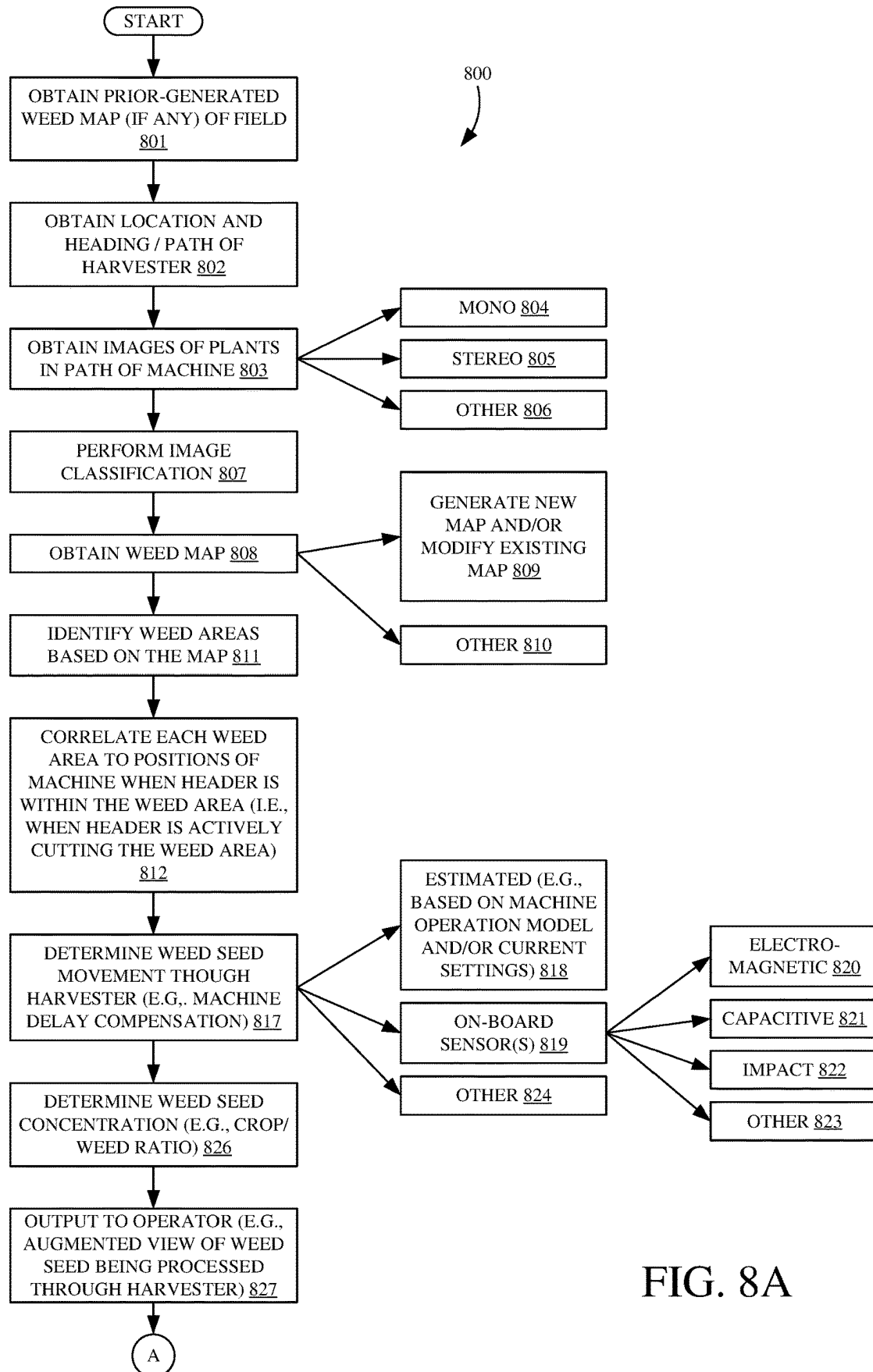


FIG. 8A

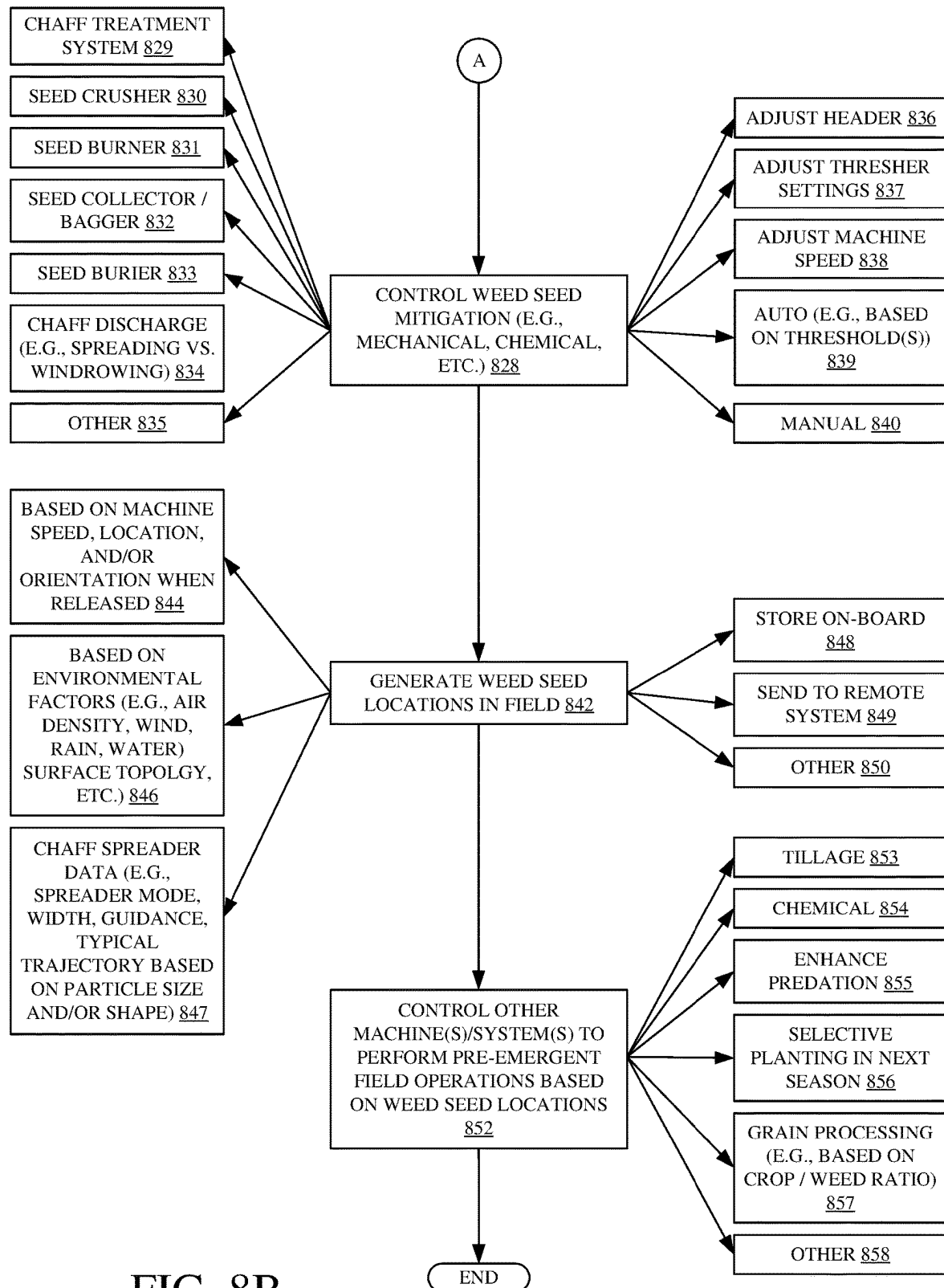
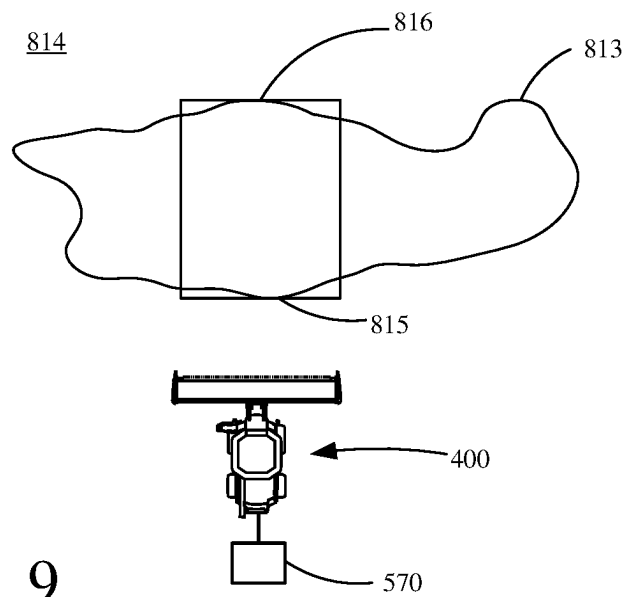


FIG. 8B



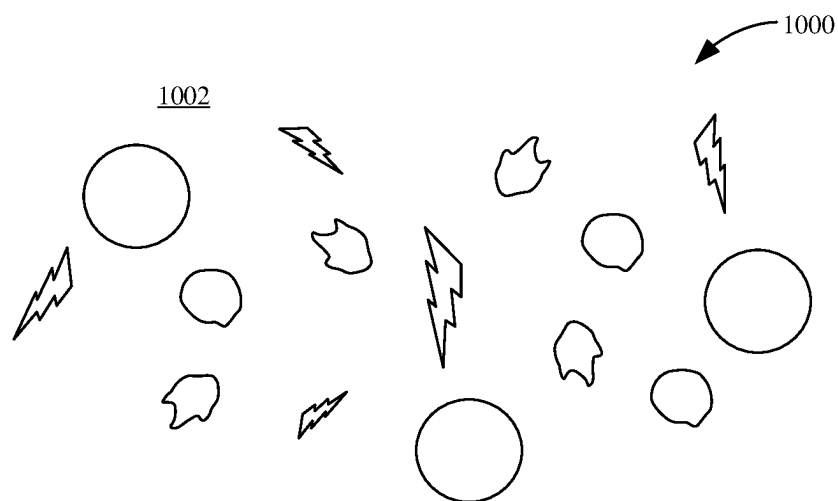


FIG. 10A

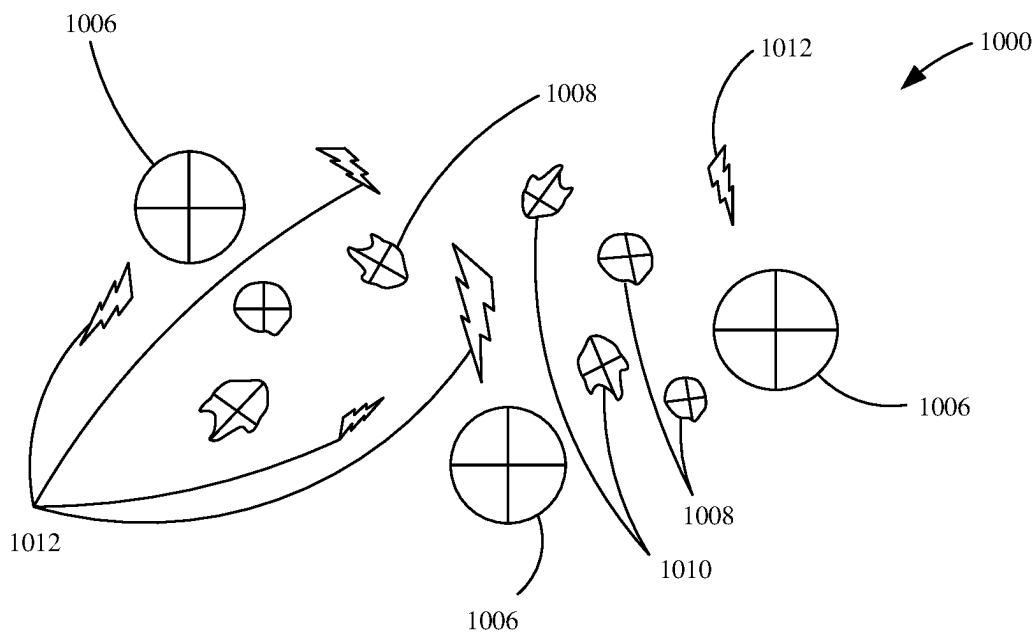


FIG. 10B

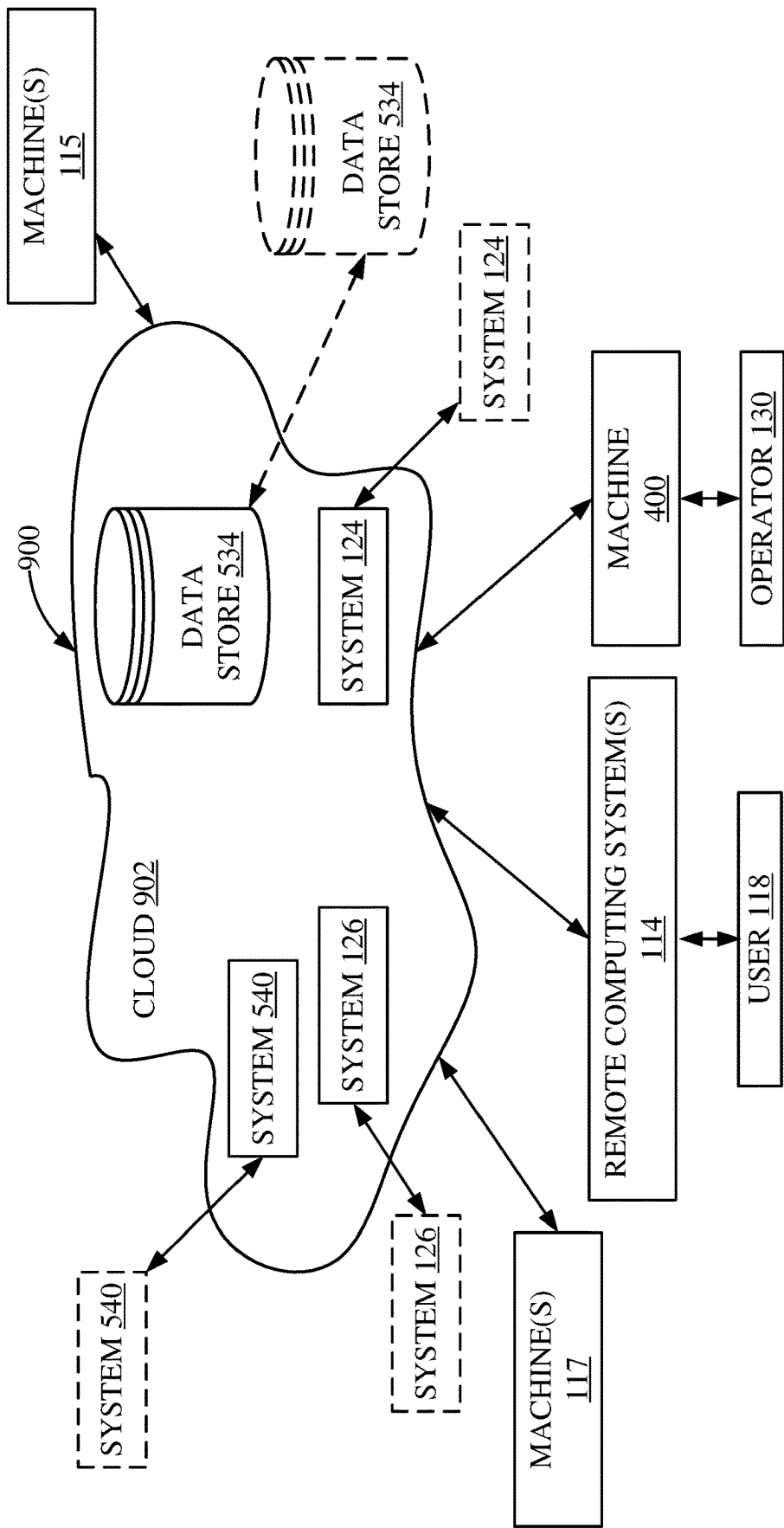


FIG. 11

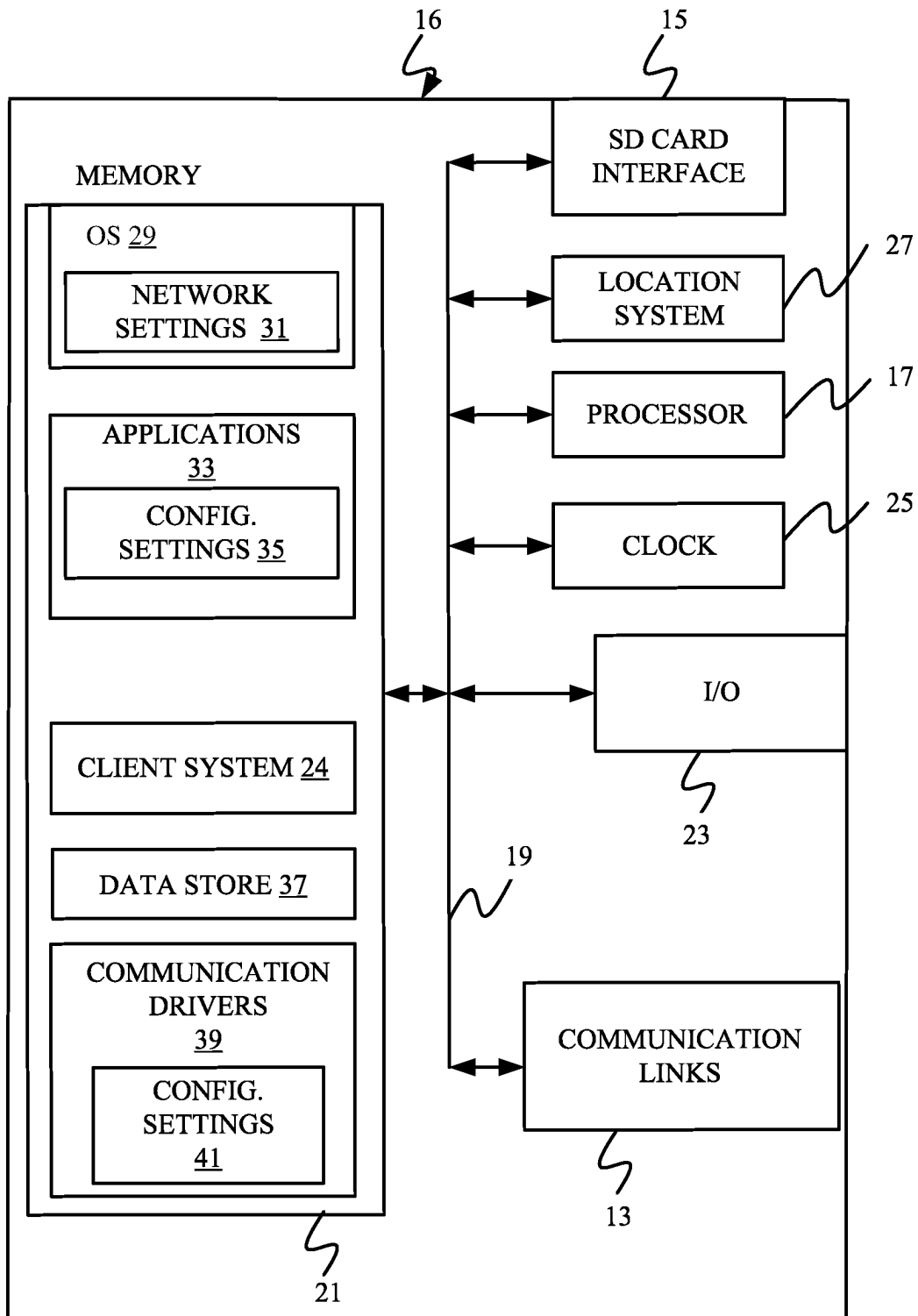


FIG. 12

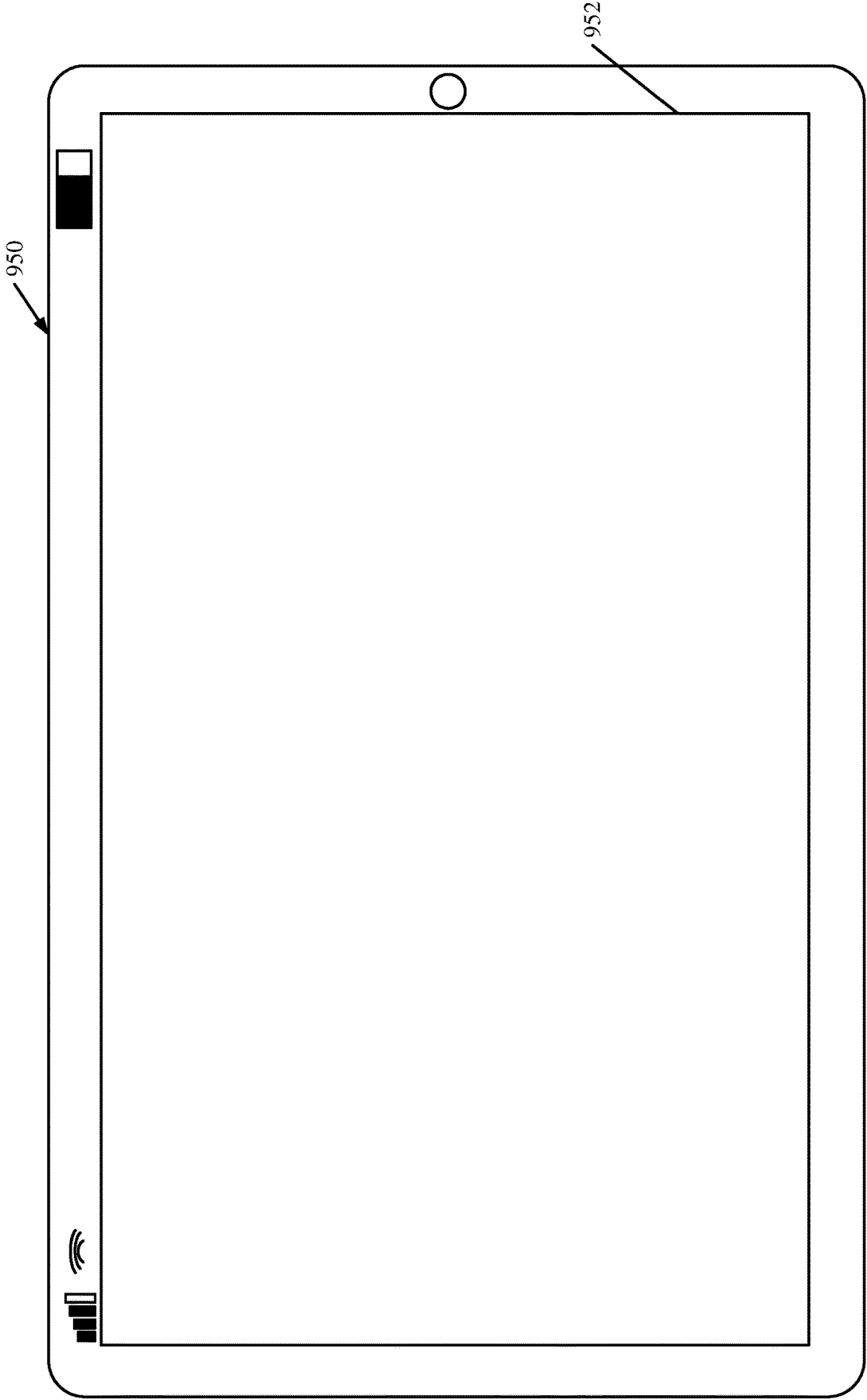


FIG. 13

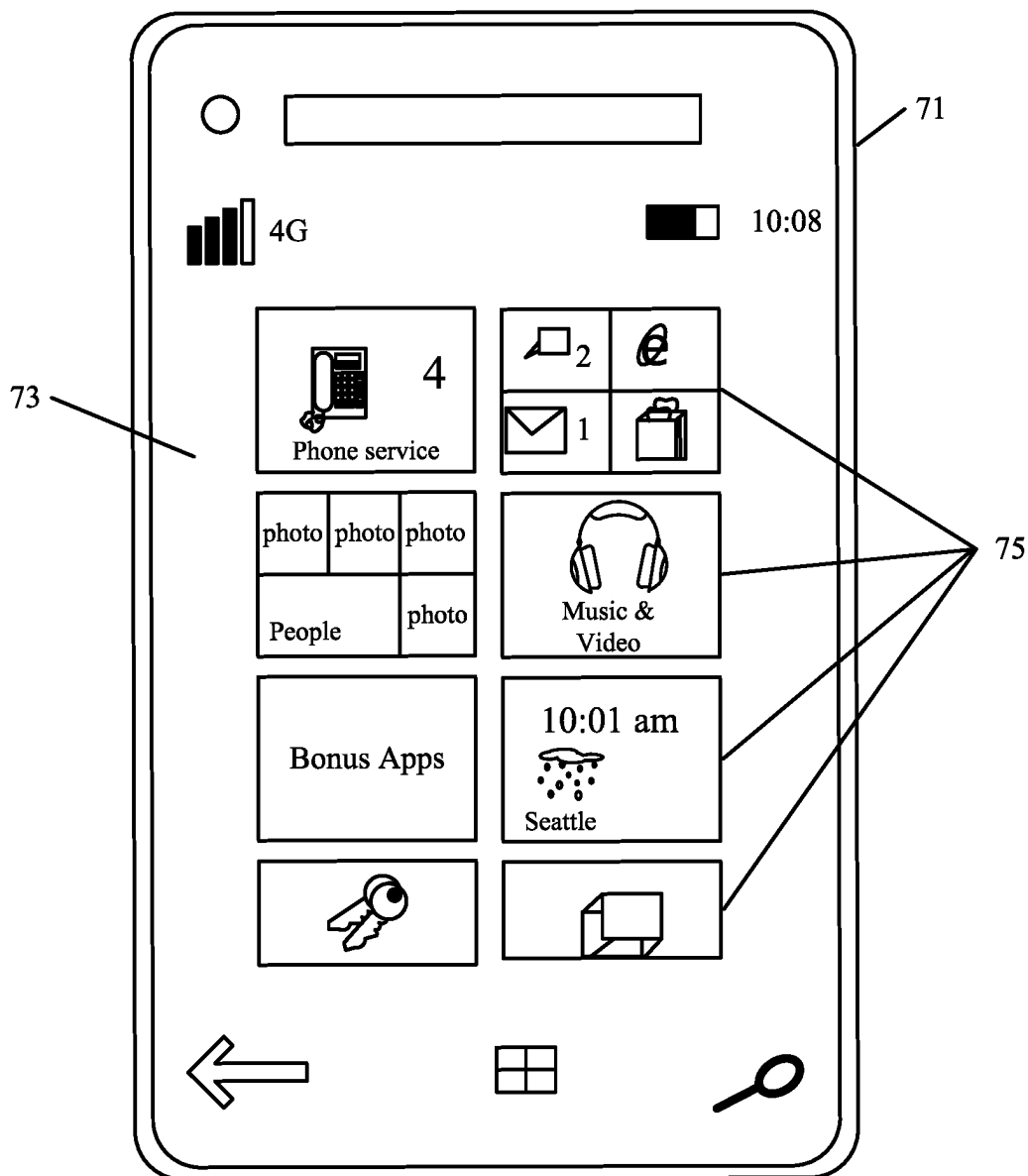


FIG. 14

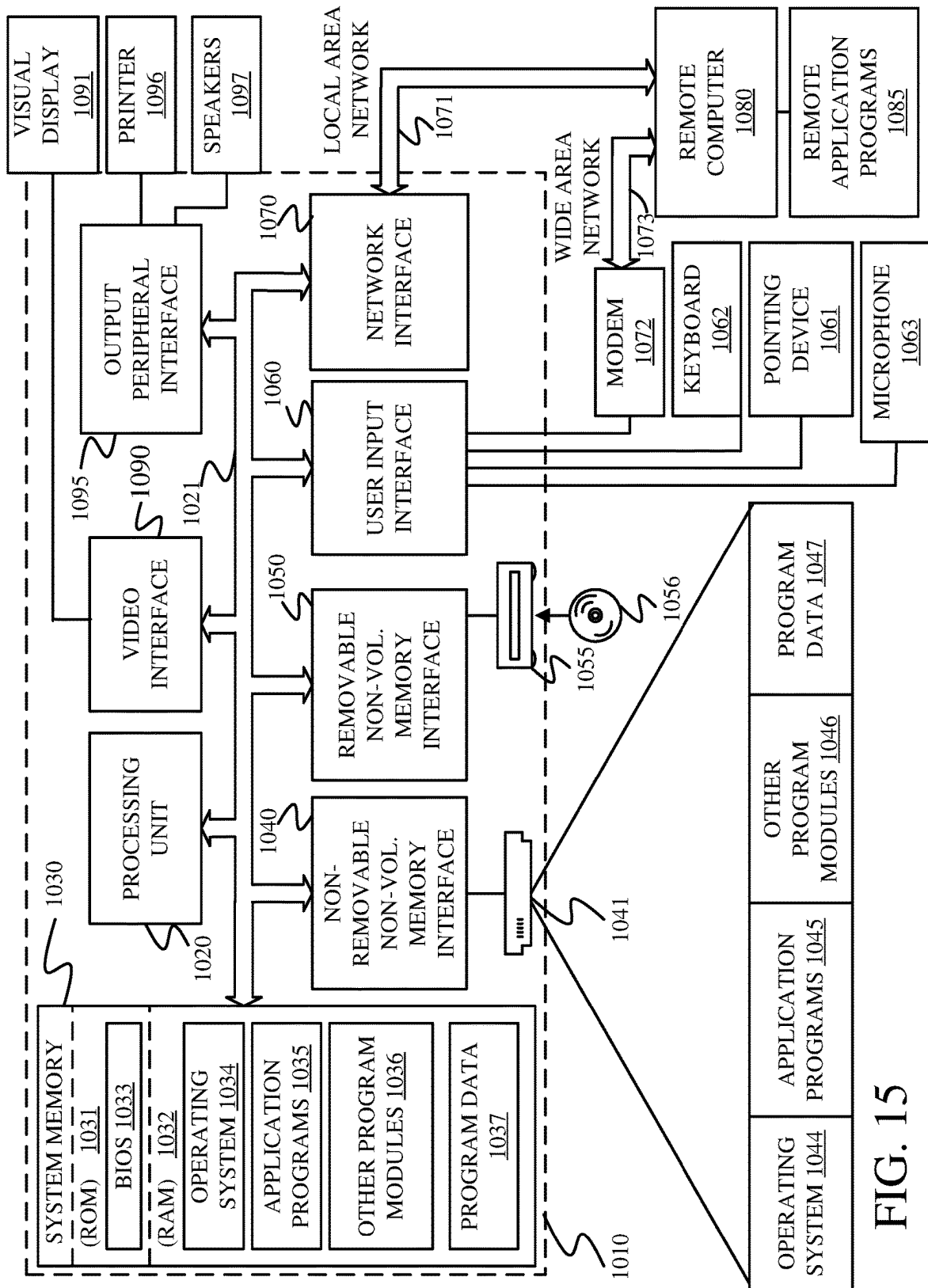


FIG. 15

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AGRICULTURAL HARVESTING MACHINE WITH PRE-EMERGENCE WEED DETECTION AND MITIGATION SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

The present application is a continuation of and claims priority of U.S. patent application Ser. No. 16/783,475, filed Feb. 6, 2020, the content of which is hereby incorporated by reference in its entirety. The present application is also a continuation of and claims priority of U.S. patent application Ser. No. 16/783,511, filed Feb. 6, 2020, the content of which is hereby incorporated by reference in its entirety.

FIELD OF THE DESCRIPTION

The present description generally relates to agricultural machines. More specifically, but not by limitation, the present description relates to pre-emergence weed detection and mitigation.

BACKGROUND

There are a wide variety of different types of farming techniques. One such technique is referred to as precision farming. Precision farming, or precision agriculture, is also referred to as site-specific crop management. The technique uses observation and measurement of variations of different criteria at specific sites, from field-to-field, and even within a single field. The observation and measurement of the variation in the different criteria can then be acted on in different ways.

The effectiveness of precision farming depends, at least in part, upon the timely gathering of information at a site-specific level, so that information can be used to make better decisions in treating and managing the crop. This type of information can include information that is indicative of plant emergence characteristics (such as maturity, emergence uniformity, etc.) pest presence, disease, water and nutrient levels, weed stresses, etc. Management techniques for weeds, which reduce crop yields, include the application of a chemical (e.g., herbicide) to the field to mitigate weed growth.

The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

An agricultural harvesting machine includes crop processing functionality configured to engage crop in a field, perform a crop processing operation on the crop, and move the processed crop to a harvested crop repository, and a control system configured to identify a weed seed area indicating presence of weed seeds, and generate a control signal associated with a pre-emergence weed seed treatment operation based on the identified weed seed area.

This Summary is provided to introduce a selection of concepts in a simplified form that is further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

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BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates one example of an agricultural architecture for pre-emergence weed mitigation.

FIG. 2 is a flow diagram illustrating an example operation of an agricultural architecture that identifies weed seed locations and performing a pre-emergence weed mitigation operation.

FIG. 3 is a flow diagram illustrating an example operation for generating weed map(s).

FIG. 4 is a partial pictorial, partial schematic, illustration of one example of an agricultural machine.

FIG. 5 is a block diagram showing one example of an agricultural machine.

FIG. 6 is a block diagram illustrating one example of a control system.

FIG. 7 is a block diagram illustrating one example of a pre-emergence weed mitigation system.

FIGS. 8A and 8B is a flow diagram illustrating an example operation of a pre-emergence weed mitigation system.

FIG. 9 illustrates an example weed area on a field being harvested by a combine.

FIGS. 10A and 10B illustrates an example image from an electromagnetic sensor that senses components of a material flow in an agricultural harvesting machine.

FIG. 11 is a block diagram showing one example of the architecture illustrated in FIG. 3, deployed in a remote server architecture.

FIGS. 12-14 show examples of mobile devices that can be used in the architectures shown in the previous figures.

FIG. 15 is a block diagram showing one example of a computing environment that can be used in the architectures shown in the previous figures.

DETAILED DESCRIPTION

The present description generally relates to agricultural machines. More specifically, but not by limitation, the present description relates to pre-emergence weed detection and mitigation. As noted above, some weed management techniques including the application of a chemical (e.g., herbicide) to an agricultural field to mitigate weed growth. For sake of the present discussion, a “weed” or “weed plant” refers to any plant other than a target crop plant type of the subject field. This can include non-crop plants as well as crop plants of a different crop type. To illustrate, in a corn field to be harvested by a corn harvester, “weeds” can include common non-crop plants (e.g., giant ragweed, common ragweed, horseweed (marestail), johnsongrass, palmer amaranth, ryegrass, waterhemp, etc.) and crop plants other than corn (e.g., soybeans, etc.). That is, it includes plant types other corn plants.

Unfortunately, over time some types of weeds have developed herbicide-resistance which results in decreased effectiveness of the herbicide application. For instance, examples of weeds that have developed glyphosate resistance include, but are not limited to, those mentioned above. At best, the herbicide-resistance requires an excessive application of the herbicide and, at worst, the herbicide-resistance renders the herbicide application ineffective. Further, excessive application of herbicide has drawbacks. For instance, in addition to a significant increase in costs involved (e.g., machine operating costs, herbicide costs, etc.), excessive herbicide application may be harmful to the crop and/or is otherwise undesirable.

One pre-emergence application technique utilizes weed maps and an expected timing of emergence to determine when to apply a pre-emergence herbicide. These maps are obtained from weed growing locations from prior year growing seasons or harvest, to predict where the weeds will emerge for the current year. This is often inaccurate, which can result in incorrect herbicide application doses and/or the application of herbicide to the incorrect areas of the field.

The present disclosure provides a system for an agricultural environment that processes weed plant location information, such as weed maps, that supports pre-emergence mitigation. The weed plant data can be obtained from any of a wide variety of sources, such as remote sensing data obtained from image data sources. Examples of image data sources include, but are not limited to, manned aircraft cameras, unmanned aerial vehicle (UAV or drone) cameras, stationary mounted or on-board cameras, etc. For sake of illustration, as discussed below an agricultural harvester or combine identifies the locations of weed seeds, which can be utilized to control on-board weed seed mitigators. Alternatively, or in addition, weed seed maps can be generated and utilized to perform pre-emergence weed mitigation post-harvest. In either case, the system can mitigate even herbicide-resistance weeds.

FIG. 1 illustrates one example of an agricultural architecture 100 for pre-emergence weed mitigation. Architecture 100 includes an agricultural machine 102 configured to generate pre-emergence weed seed location information that represents the presence of weed seeds in a field and/or perform a pre-emergence weed mitigation operation using that weed seed location information. It is noted that machine 102 can be any of a wide variety of different types of agricultural machines. For instance, in examples described below machine 102 comprises an agricultural harvesting machine (also referred to as a “harvester” or “combine”). In other examples, machine 102 can comprise a sprayer, cultivator, to name a few. Also, while machine 102 is illustrated with a single box in FIG. 1, machine 102 can comprise multiple machines (e.g., a towed implement towed by a towing machine). In this example, the elements of machine 102 illustrated in FIG. 1 can be distributed across a number of different machines.

Machine 102 includes a control system 108 configured to control other components and systems of architecture 100. For instance, control system 108 includes a weed seed mapping system 109, which is discussed in further detail below. Also, control system 108 includes a communication controller 110 configured to control communication system 112 to communicate between components of machine 102 and/or with other machines or systems, such as remote computing system 114 and/or machine(s) 115, either directly or over a network 116. Also, machine 102 can communicate with other agricultural machine(s) 117 as well. Agricultural machine(s) 117 can be a similar type of machine as machine 102, and they can be different types of machines as well. Network 116 can be any of a wide variety of different types of networks such as the Internet, a cellular network, a local area network, a near field communication network, or any of a wide variety of other networks or combinations of networks or communication systems.

A remote user 118 is illustrated interacting with remote computing system 114. Remote computing system 114 can be a wide variety of different types of systems. For example, remote system 114 can be a remote server environment, remote computing system that is used by remote user 118. Further, it can be a remote computing system, such as a mobile device, remote network, or a wide variety of other

remote systems. Remote system 114 can include one or more processors or servers, a data store, and it can include other items as well.

Communication system 112 can include wired and/or wireless communication logic, which can be substantially any communication system that can be used by the systems and components of machine 102 to communicate information to other items, such as between control system 108, sensors 120, controllable subsystems 122, image capture system 124, and plant evaluation system 126. In one example, communication system 112 communicates over a controller area network (CAN) bus (or another network, such as an Ethernet network, etc.) to communicate information between those items. This information can include the various sensor signals and output signals generated by the sensor variables and/or sensed variables.

Control system 108 includes a user interface component 127 configured to control interfaces, such as operator interface(s) 128 that include input mechanisms configured to receive input from an operator 130 and output mechanisms that render outputs to operator 130. The user input mechanisms can include mechanisms such as hardware buttons, switches, joysticks, keyboards, etc., as well as virtual mechanisms or actuators such as a virtual keyboard or actuators displayed on a touch sensitive screen. The output mechanisms can include display screens, speakers, etc.

Sensor(s) 120 can include any of a wide variety of different types of sensors. In the illustrated example, sensors 120 include position sensor(s) 132, speed sensor(s) 134, environmental sensor(s) 136, and can include other types of sensors 138 as well. Position sensor(s) 132 are configured to determine a geographic position of machine 102 on the field, and can include, but are not limited to, a Global Navigation Satellite System (GNSS) receiver that receives signals from a GNSS satellite transmitter. It can also include a Real-Time Kinematic (RTK) component that is configured to enhance the precision of position data derived from the GNSS signal. Speed sensor(s) 134 are configured to determine a speed at which machine 102 is traveling the field during the spraying operation. This can include sensors that sense the movement of ground-engaging elements (e.g., wheels or tracks) and/or can utilize signals received from other sources, such as position sensor(s) 132.

Control system 108 includes control logic 140, and can include other items 142 as well. As illustrated by the dashed box in FIG. 1, control system 108 can include some or all of plant evaluation system 126, which is discussed in further detail below. Also, machine 102 can include some or all of image capture system 124. Control logic 140 is configured to generate control signals to control sensors 120, controllable subsystems 122, communication system 112, or any other items in architecture 100. Controllable subsystems 122 include a pre-emergence weed mitigation system 144, machine actuators 146, a propulsion subsystem 148, a steering subsystem 150, and can include other items 152 as well.

Machine 102 includes a data store 161 configured to store data for use by machine 102, such as field data. Examples include, but are not limited to, field location data that identifies a location of the field to be operated upon by a machine 102, field shape and topography data that defines a shape and topography of the field, crop location data that is indicative of a location of crops in the field (e.g., the location of crop rows), or any other data. In the illustrated example, data store 161 stores weed maps 162 that are generated by

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machine **102** or otherwise obtained by machine **102**, such as from plant evaluation system **126**. Of course, data store **161** can store other data as well.

Machine **102** is illustrated as including one or more processors or servers **163**, and it can include other items **164** as well.

As illustrated by the dashed boxes in FIG. 1, machine **102** can include some or all components of image capture system **124** and/or plant evaluation system **126**, both of which are discussed in further detail below. Also, agricultural machine(s) **117** can include a pre-emergence weed mitigation system **119**, which can be similar to, or different from, system **144**.

Image capture system **124** includes image capture components configured to capture one or more images of the area under consideration (i.e., the portions of the field to be operated upon by machine **102**) and image processing components configured to process those images. The captured images represent a spectral response captured by image capture system **124** that are provided to plant evaluations system **126** and/or stored in data store **174**. A spectral imaging system illustratively includes a camera that takes spectral images of the field under analysis. For instance, the camera can be a multispectral camera or a hyperspectral camera, or a wide variety of other devices for capturing spectral images. The camera can detect visible light, infrared radiation, or otherwise.

In one example, the image capture components include a stereo camera configured to capture a still image, a time series of images, and/or a video of the field. An example stereo camera captures high definition video at thirty frames per second (FPS) with one hundred and ten degree wide-angle field of view. Of course, this is for sake of example only.

Illustratively, a stereo camera includes two or more lenses with a separate image sensor for each lens. Stereo images (e.g., stereoscopic photos) captured by a stereo camera allow for computer stereo vision that extracts three-dimensional information from the digital images. In another example, a single lens camera can be utilized to acquire images (referred to as a "mono" image).

Image capture system **124** can include one or more of an aerial image capture system **176**, an on-board image capture system **178**, and/or other image capture system **180**. An example of aerial image capture system **124** includes a camera or other imaging component carried on an unmanned aerial vehicle (UAV) or drone (e.g., block **115**). An example of on-board image capture system **178** includes a camera or other imaging component mounted on, or otherwise carried by, machine **102** (or **104**). An example of image capture system **180** includes a satellite imaging system. System **124** also includes a location system **182**, and can include other items **184** as well. Location system **182** is configured to generate a signal indicative of geographic location associated with the captured image. For example, location system **182** can output GPS coordinates that are associated with the captured image to obtain geo-referenced images **186** that are provided to plant evaluation system **126**.

Plant evaluation system **126** illustratively includes one or more processors **188**, a communication system **189**, a data store **190**, an image analysis system **191**, target field identification logic **192**, trigger detection logic **193**, a weed map generator **194**, and can include other items **195** as well. Communication system **189**, in one example, is substantially similar to communication system **112**, discussed above.

Target field identification logic **192** is configured to identify a target or subject field for which a weed map is to be generated by weed map generator **194**. The target field

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identification is correlated to the weed maps **196**, which are generated by weed map generator **194** and can be stored in data store **190**.

Trigger detection logic **193** is configured to detect a triggering criterion that triggers generation (or updating) of a weed map by generator **194**. For example, in response to detection of a triggering criteria, logic **193** can communicate instructions to image capture system **124** to capture images of the target field. These images are then processed by image analysis system **191**, and the results of the image analysis are utilized by weed map generator **194** to generate weed maps **196**.

FIG. 2 is a flow diagram **200** illustrating an example operation of architecture **100** in identifying weed seed locations and performing a pre-emergence weed mitigation operation.

At block **202**, logic **192** identifies a target worksite (i.e., a field to be harvested). At block **204**, logic **193** detects a trigger for triggering generation (or updating) of a weed map for the identified field. For instance, this can be done periodically (block **206**), in response to an event (block **208**), and/or manually in response to a user input (block **210**). Of course, the trigger can be detected in other ways as well. This is represented by block **212**.

At block **214**, a weed map of the field is generated. It is noted that the weed map can be generated at any of a variety of different times. For example, the weed map can be generated during the growing season, before harvest while the crops (and weeds) are growing. This is represented by block **216**. In another example, the weed map can be generated at harvest time, when a harvesting machine is performing a harvesting operation in the field. This is represented by block **218**. In another example, the weed map can be generated by a combination of inputs during the growing season and at harvest time. This is represented by block **220**. Of course, the weed map can be generated in other ways as well. This is represented by block **222**. In one example, the weed map can include two (or more) plant classifications, i.e., crop and weed. Alternatively, or in addition, the weed map can include multiple non-crop plant classifications based on, for example but not by limitation, species, size, maturity, vigor, etc.

In one example, image capture system **124** captures spectral images of the field under analysis, as well as video images. Geographic location information is associated with those images, and they are provided to plant evaluation system **126**. System **126** identifies evaluation zones in the field under analysis and analyzes the spectral images received from system **124** to identify weed plants in the evaluation zones. This can be done in any of a number of ways. For instance, the images can be processed to identify areas that correspond to weed plants. In another example, system **126** can identify areas in the evaluation zones that represent crop plants and subtract those areas out of the images to obtain a remaining image portion that represents the weeds or non-crop plants.

In one example, the image capture system includes a camera, such as a multispectral camera or a hyperspectral camera, or a wide variety of other devices for capturing images. A video imaging system can be utilized that includes a camera that captures images in the visible or thermal image range. For example, it can be a visible light video camera with a wide angle lens, or a wide variety of other video imaging systems.

Additionally, plant density information can be generated and associated with the weed map. That is, in addition to the weed map identifying areas of the field that contain weeds,

a density metric can be associated with those areas. For instance, the density metric can indicate a percentage of the plants within the area that are weed plants versus crop plants. In another instance, it can be weeds/unit area.

In one example, image analysis system **191** includes spectral analysis logic that performs spectral analysis to evaluate the plants in the images. In one example, this includes identifying areas in the image that have a spectral signature that corresponds to ground versus plants. For instance, this can be a green/brown comparison. Image segmentation logic can perform image segmentation to segment or divide the image into different portions for processing. This can be based on ground and/or plant area identifications by ground/plant identification logic, and crop classification performed by crop classification logic. Briefly, this can identify areas of an image that represent ground and areas of an image that represent plants, for example using the spatial and spectral analysis. Crop classification logic can use a crop classifier, that is trained using crop training data, to identify areas in the image that represent crop plants and/or areas that represent weed plants.

In addition to identifying the location of the plant relative to the surface plane of the field (e.g., x/y coordinates), a height of the weed plants can be identified (e.g., how high the plant rises from the terrain in the z direction).

At block **224**, weed seed locations are identified. The weed seed locations identify the location of the weed seeds pre-emergence, that is before the seeds germinate and emerge as visible plants. The weed seed locations can be identified in any of a number of ways. For example, the weed seed locations can be identified based on a priori data (block **226**), in situ data (block **228**), or a combination of a priori and in situ data (block **230**). For instance, the weed seed locations can be based on an a priori weed map generated during the growing season at block **216**. Alternatively, or in addition, the weed seed locations can be identified based on in situ data collected by on-board sensors.

As illustrated at block **232**, the identified weed seed locations can be utilized to generate a weed seed map that maps the locations of the weed seeds to the field under analysis. An example weed seed map identifies regions of the field that are determined to have a number of weed seeds above a threshold, which can be defined in any of a number of ways. For example, the threshold can be pre-defined, set by an operator, dynamically determined, etc.

As illustrated at block **234**, the weed seed locations are identified based on the weed map, generated at block **294**, which maps locations of the weeds in the field, taking into account a weed seed movement model. This model projects the likely location of a weed plant's seeds given the location of that weed plant and external factors that affect movement of the seed from the weed plant location. For instance, the model can take into account weather or other environmental data. For instance, the location of the weed seeds on the field can be determined based on the direction and/or speed of the wind as detected by sensors on machine **102** or otherwise obtained from a remote weather data source. In another example, the weed seed model can identify terrain conditions, such as slope or topography, precipitation, among other factors which can contribute to the displacement of the seeds from a weed plant.

Alternatively, or in addition, the weed seed movement model can model machine data that processes the weed plants. For example, in the case of an agricultural harvesting machine, the machine data can be utilized to compensate for machine delays caused by the processing through the com-

bine. That is, the machine delay models the distance (with respect to the field surface) between when the weed plant is cut by the header of the combine and the weed seeds are discharged by a chaff spreader. This delay can be dynamically determined based on machine settings (header speed, thresher settings, chaff settings, etc.) that may vary the time that it takes for the seed to travel through the combine and be discharged onto the field. As used herein, chaff refers to any material (also referred to a "residue") that is ejected from the harvesting machine (typically from the rear of the machine), even though it may contain some crop material. That is, during operation of the combine, it is often the case that some crop material ends up in the non-crop material flow, and vice versa. Of course, the weed seed locations can be identified in other ways as well. This is represented by block **236**.

At block **238**, the current weed seed locations (e.g., the weed seed map generated at block **232**) is stored. The weed seed locations can be stored locally (e.g., in data store **161**), can be sent to another agricultural machine (e.g., machine **117**), and/or can be sent to a remote computing system (e.g., system **114**).

At block **240**, a control signal is generated for a pre-emergence weed mitigation operation. This can be performed during and/or after a harvesting operation. For example, a mitigation operation performed during the harvesting operation comprises a selective harvest. This is represented by block **242**. For instance, the harvesting machine can be controlled to selectively harvest different areas of the field based on the weed seed locations. That is, an area of high weed seed occurrence can be ignored, and then mitigated after the harvesting operation. In another example, the harvesting operation can selectively harvest areas of high weed seed presence in a single harvesting operation (so all of the material is collected together in the material repository) and then can be subsequently processed. These, of course, are for sake of example only.

In another example, the weed seeds are collecting during the harvesting operation. For instance, a collector or other apparatus is positioned to collect the discharge from the combine and prevent the weed seeds from being ejected back onto the field. Alternatively, or in addition, a mitigator can be utilized to destroy or otherwise devitalize the weed seeds, inhibiting further germination or promulgation of the weed seeds. This can include mechanical mitigators, chemical mitigators, irradiation mitigators, etc. Examples of this are discussed in further detail below. Briefly, however, an example mitigator (mechanical, chemical, or otherwise) includes a device that interacts with the weed seed such that the weed seed has a lower ability to promulgate or germinate in a subsequent growing season.

Also, the pre-emergence weed mitigation operation can discourage growth of the weed seeds. This is represented by block **248**. For example, a tiller machine can be utilized to till the area, post-harvest, to bury the weed seeds at a threshold depth (e.g., twelve inches or greater) at which the weed seeds are unlikely to germinate. In another example, early germination of the weed seeds can be stimulated (i.e., during the fall) so that the germinated weeds are exposed to the cold fall/winter weather which is likely to destroy the weed plants. In another example, a chemical can be applied to the weed seeds to discourage their spring germination and/or increase predation (e.g., being consumed by predator animals).

Of course, the pre-emergence weed mitigation operation can comprise other types of operations as well. This is represented by block **250**.

FIG. 3 is a flow diagram 300 illustrating an example operation for generating weed map(s). For sake of illustration, but not by limitation, FIG. 3 will be described in the context of systems 124 and 126 generating weed maps 196 for use by machine 102.

A block 302, image data of the field is collected during the growing season and/or at harvest. As discussed above, this can include multispectral and/or hyperspectral images, represented by block 304. Alternatively, or in addition, closeup video of the plants can be obtained at block 306. Of course, other images can be obtained as well. This is represented by block 308.

Also, the image data can be collected from a wide variety of different sources, for example, the image data can be collected from a UAV (block 310), a satellite system (block 312), on-board cameras (block 314) that are on board machine 102, and/or from other machines or devices (block 316).

A physiological plant growth model can be obtained at block 318. Illustratively, a plant growth model can be used to understand what weed/crop maturity stage(s) to expect at a given time and location in the field. This can facilitate improvement of classifier results, especially if the characteristics change significantly during the growth cycle (i.e. less misclassification, better ability to differentiate). The model can represent irrigation patterns of the field (block 320), weather data (block 322), and/or soil/terrain data (block 324). The weather data at block 322 can represent precipitation during the growing season and the soil/terrain data 324 can indicate soil characteristics, such as moisture, etc., as well as terrain topology data, such as the slope of the field.

Also, the plant growth model can be generated based on data from a farm management information system (FMIS). This is represented by block 326. An example FMIS system provides information on the type and/or variety of the planted crop, plant date of the crop, treatments that have been applied to the crop (e.g., before or during the growing season). Of course, the model can be obtained using other data as well. This is represented by block 328.

At block 330, crop characteristics are obtained. In one example, this models where the crop maturity should be given the growth model. For instance, the crop characteristics obtained at block 330 can indicate that the crop and/or weeds should be at a particular emergence stage. In one example, this can represent crop/weed dry up characteristics prior to harvest. This can be utilized to differentiate crops and weeds, and is represented by block 332. In another example, relative maturity rate data can be obtained at block 334. In one example, this is utilized to find zones of similar agronomic behavior that can be utilized to classify weeds versus crops in smaller areas of the field, rather than across the entire field collectively.

Of course, other crop characteristics can be obtained as well. This is represented by block 336. At block 338, the images collected at block 332 are classified to obtain a georeferenced weed map. For example, the classification can utilize a normalized difference vegetation index (NDVI). In one example, a crop mask is applied on the NDVI to obtain better crop development monitoring.

In another example, weed/crop identifying characteristics can include reflectance spectra (block 342), leaf shape information (block 344), population texture (block 346), plant height (block 348), etc. Of course, the images can be classified in other ways as well. This is represented by block 350.

At block 352, the weed map is output. For example, the weed map can be output to control system 108 for use by weed seed mapping system 109 to control pre-emergence weed mitigation system 144. In another example, the weed map can be output for storage and/or display to remote user 118 and/or operator 130 using a display device.

As noted above, one example of agricultural machine 102 is a harvester or combine, such as that shown in FIG. 4 which is a partial pictorial, partial schematic, illustration of an agricultural harvesting machine 400 (or combine). It can be seen in FIG. 4 that combine 400 illustratively includes an operator compartment 401, which can have a variety of different operator interface mechanisms, for controlling combine 400, as will be discussed in more detail below. Combine 400 can include a set of front end equipment that can include header 402, and a cutter generally indicated at 404. It can also include a feeder house 406, a feed accelerator 408, and a thresher generally indicated at 410. Thresher 410 illustratively includes a threshing rotor 412 and a set of concaves 414. Further, combine 400 can include a separator 416 that includes a separator rotor. Combine 400 can include a cleaning subsystem (or cleaning shoe) 418 that, itself, can include a cleaning fan 420, chaffer 422 and sieve 424. The material handling subsystem in combine 400 can include (in addition to a feeder house 406 and feed accelerator 408) discharge beater 426, tailings elevator 428, clean grain elevator 430 (that moves clean grain into clean grain tank 432) as well as unloading auger 434 and spout 436. Combine 400 can further include a residue subsystem 438 that can include chopper 440 and spreader 442. Combine 400 can also have a propulsion subsystem that includes an engine that drives ground engaging wheels 444 or tracks, etc. It will be noted that combine 400 may also have more than one of any of the subsystems mentioned above (such as left and right cleaning shoes, separators, etc.).

In operation, and by way of overview, combine 400 illustratively moves through a field in the direction indicated by arrow 446. As it moves, header 402 engages the crop to be harvested and gathers it toward cutter 404. After it is cut, it is moved through a conveyor in feeder house 406 toward feed accelerator 408, which accelerates the crop into thresher 410. The crop is threshed by rotor 412 rotating the crop against concave 414. The threshed crop is moved by a separator rotor in separator 416 where some of the residue is moved by discharge beater 426 toward the residue subsystem 438. It can be chopped by residue chopper 440 and spread on the field by spreader 442. In other implementations, the residue is simply dropped in a windrow, instead of being chopped and spread.

Grain falls to cleaning shoe (or cleaning subsystem) 418. Chaffer 422 separates some of the larger material from the grain, and sieve 424 separates some of the finer material from the clean grain. Clean grain falls to an auger in clean grain elevator 430, which moves the clean grain upward and deposits it in clean grain tank 432. Residue can be removed from the cleaning shoe 418 by airflow generated by cleaning fan 420. That residue can also be moved rearwardly in combine 400 toward the residue handling subsystem 438.

Tailings can be moved by tailings elevator 428 back to thresher 410 where they can be re-threshed. Alternatively, the tailings can also be passed to a separate re-threshing mechanism (also using a tailings elevator or another transport mechanism) where they can be re-threshed as well.

FIG. 4 also shows that, in one example, combine 400 can include ground speed sensor 447, one or more separator loss sensors 448, a clean grain camera 450, one or more cleaning shoe loss sensors 452, forward looking camera 454, rear-

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ward looking camera **456**, a tailings elevator camera **458**, and a wide variety of other cameras or image/video capture devices. Ground speed sensor **446** illustratively senses the travel speed of combine **400** over the ground. This can be done by sensing the speed of rotation of the wheels, the drive shaft, the axel, or other components. The travel speed can also be sensed by a positioning system, such as a global positioning system (GPS), a dead reckoning system, a LORAN system, or a wide variety of other systems or sensors that provide an indication of travel speed. In one example, optical sensor(s) capture images and optical flow is utilized to determine relative movement between two (or more) images taken at a given time spacing.

Cleaning shoe loss sensors **452** illustratively provide an output signal indicative of the quantity of grain loss. In one example, this includes signal(s) indicative of the quality of grain loss by both the right and left sides of the cleaning shoe **418**. In one example, sensors **452** are strike sensors which count grain strikes per unit of time (or per unit of distance traveled) to provide an indication of the cleaning shoe grain loss. The strike sensors for the right and left sides of the cleaning shoe can provide individual signals, or a combined or aggregated signal. In one example, sound-based sensors across an area of the cleaning shoe and/or rotor can be utilized to obtain a count of grain strikes and a spatial distribution associated with the count. It will be noted that sensors **452** can comprise only a single sensor as well, instead of separate sensors for each shoe.

Separator loss sensor **448** provides a signal indicative of grain loss in the left and right separators. The sensors associated with the left and right separators can provide separate grain loss signals or a combined or aggregate signal. This can be done using a wide variety of different types of sensors as well. It will be noted that separator loss sensors **448** may also comprise only a single sensor, instead of separate left and right sensors.

Cameras **450**, **454**, **456** and **458** illustratively capture video or still images that can be transmitted to, and displayed on, a display in operator compartment **401** or a remote device (shown in more detail below) in near real time. Clean grain camera **450**, for instance, generates a video feed showing grain passing into clean grain tank **432** (or through clean grain elevator **430**). Camera **454** can illustratively generate a video feed showing a view forward of operator compartment **401**, such as showing header **402** and/or the crop in front of header **402**. Cameras **456** and **458** illustratively generate a video feed showing the tailings in elevator **458** and the discharge beater **442** and an area of the field behind combine **400**, respectively. Alternatively, or in addition to a video feed, captured images can be augmented and presented to the operator, for example in a manner aimed to reduce cognitive load on the operation. These are examples only, and additional or different cameras can be used and/or they can be devices that capture still images or other visual data.

FIG. **5** is a block diagram showing one example of machine **400** deployed in architecture **100**. As illustrated, combine **400** is configured to communicate over network **116** with remote computing system(s) **114**, as well as a set of assistive machines **502**, which can include a chaff collector **504**, a grain cart **506**, a tiller **508**, and/or other machines **510**. Machine **400** can also communicate with one or more grain receivers/processors **512**. This can include a grain truck **514**, a grain bin **516**, a cleaner **518**, an incinerator **520**, a burner **522**, and it can include other items **524** as well.

As shown in FIG. **5**, combine **400** can generate operator interface displays **526** with user input mechanisms **528** for

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interaction by operator **130**. Operator **130** is illustratively a local operator of combine **400**, in the operator's compartment **401**, and can interact with user input mechanisms **528** in order to control and manipulate combine **400**.

Combine **400** includes one or more processors or servers **530**, communication system **532**, a data store **534**, sensors **536**, controllable subsystems **538**, a control system **540**, a user interface component **542**, and can include other items **544** as well. In one example, some or all of these components are similar to the components discussed above with respect to machine **102** shown in FIG. **1**.

User interface component **542** can include one or more display devices, audio devices, one or more haptic devices, and it can include other items, such as a steering wheel, one or more joysticks, pedals, levers, buttons, keypads, etc. Where the user interface mechanisms include a user interface display, then user input mechanisms **528** can include buttons, icons, actuatable links, or other items that can be actuated by operator **130**. When control system **540** or other items on machine **400** use speech recognition, and/or speech synthesis, then user interface mechanisms **528** can include a microphone, a speaker, etc.

Sensors **536**, in one example, includes one or more sensors **120** discussed above. That is, sensors **536** can include speed sensor(s) **546**, position sensor(s) **548**, environmental sensor(s) **550**, and can include other sensor(s) **552** as well. In the illustrated example, combine **400** includes weed seed sensor(s) **554** configured to sense the presence of weed seeds within combine **400**. This is discussed in further detail below. Briefly, however, sensors **554** can comprise a wide variety of different types of sensors. For instance, sensors **554** can include electromagnetic sensors, capacitive sensors, impact sensors, to name a few. In any case, sensors **554** are configured to detect the presence of weed seeds that are distinguished from crop or other material being processed through combine **400**.

Control system **540** can include logic and actuators or other items that can perform various types of processing and generate control signals to control controllable subsystems **538**. The control signals can be generated based on user inputs, they can be generated automatically based on sensor inputs, based on detected events or otherwise. They can also be generated based on remote control inputs received from remote computing system **114**.

Controllable subsystems **538** illustratively includes thresher **556**, which includes rotor **412**, concave(s) **414**, etc. Also, controllable subsystems **538** can include cleaner **558**, which includes cleaning shoe and/or fan **420**, sieve **424**, etc. Subsystems **538** also include chaff processing system **560**, which includes chaff separator **562**, chaff spreader **564**, etc. Controllable subsystems **538** also includes header **402**, a propulsion system **566** (e.g., system **148**), steering system **568** (e.g., system **150**), pre-emergence weed mitigation system **570** (e.g., system **144**), and can include other items **572** as well.

It is noted that, in one example, machine **400** is deployed in a fleet of harvesting machines (which can be the same and/or different than machine **400**) that harvest a field or group of fields. The fleet of harvesting machines can be paired with one or more mitigators (e.g., **502**, **512**) that perform weed seed mitigation operations (e.g., seed collection, devitalization, etc.) for the entire fleet of machines. Accordingly, each harvesting machine can send weed seed location information to the mitigator(s) (or to another system that is accessed by the mitigator(s)) which then perform the mitigation operations in the field(s).

FIG. 6 is a block diagram illustrating one example of control system 540. System 540 illustratively includes target field identification logic 602, weed map identification logic 604, a weed seed mapping system 606, a user interface component 608, control logic 610, a communication controller 612, one or more processors 614, and can include other items 616 as well. In the illustrated example, control system 540 can also include weed map generator 194, discussed above with respect to FIG. 1.

Control system 540 is illustrated as receiving a number of inputs including, but not limited to, weed maps 618 and sensor signal(s) 620. Sensor signal(s) 620 can include geo-position sensor signals 622, speed signals 624, images 626, machine delay compensation signals 628, weed seed data 630, environmental data 632, chaff spreader data 634, and can include other sensor signals as well (represented by block 636). Control system 540 is also illustrated as generated a number of outputs including, but not limited to, control signal(s) 638, weed seed location 640 (which can be stored in data store 534), and user interface mechanisms 642.

Target field identification logic 602 is configured to identify the field under consideration. This can be done based on user input from operator 130, remote user 118, or otherwise. Weed map identification logic 604 is configured to identify the weed map for the corresponding field identified by logic 602. For instance, logic 604 can receive weed maps 618 generated external to control system 540. In another example, the weed maps identified by logic 604 can be generated by weed map generator 194 on control system 540.

Weed seed mapping system 606 includes weed area identification logic 644 configured to identify weed areas based on the weed map, and includes weed seed location identification logic 646 configured to identify those locations on the field. For example, this can include a weed seed map generator 648 generating a weed seed map that maps the locations of weed seeds (and can include corresponding density information) to locations in the field. Logic 646 illustratively includes weed seed movement model application logic configured to apply a movement model that models movement of weed seeds to the weed seed areas. As described herein, this can be based on environmental data (e.g., weather, etc.), terrain data (e.g., slope, soil conditions, etc.), and/or machine data (e.g., chaff spreader settings, machine speed, etc.).

In one example, machine delay compensation logic 650 is configured to compensate for delays in the processing of the plant material in machine 400, in generating the weed seed locations from the weed maps. For instance, this can include weed seed tracking logic 652 that tracks movements of the seed within machine 400. For instance, weed seed data 630 can be received from weed seed sensors that detect the weed seeds and their movement through machine 400. In one example, logic 652 includes crop/weed ratio determination logic 652 configured to detect the ration of weed seeds to crop material being harvested.

System 606 also includes machine position correlation logic 656 configured to correlate the position of machine 400 to the weed seed areas to generate weed seed location 640 and/or control signals 638.

FIG. 7 is a block diagram illustrating one example of pre-emergence weed mitigation system 570. In the illustrated example, system 570 includes one or more weed seed mitigators 702. Illustratively, mitigator 702 includes a chaff treatment system 704 having weed seed crusher(s) 706,

weed seed burner(s) 708, weed seed separators 710, and/or chemical treatment mechanisms 712, and can include other types of mitigators as well.

Weed seed crusher(s) 706 are configured to mechanically contact (e.g., crush) the weed seeds to devitalize the weed seeds which prevents, or at least discourages, germination of the weed seeds. Similarly, weed seed burner(s) 708 are configured to thermally heat the weed seeds to a temperature that destroys the weed seeds. Chemical treatment mechanisms 712 are configured to chemically treat the weed seeds. Also, an irradiation device can irradiate the weed seeds.

Mitigators 702 can also include on-board collectors and/or baggers 714 that are configured to collect the weed seeds. In one example, collectors 714 operate some or all of the chaff being ejected from combine 400, which would otherwise be placed on the ground.

In any case, collectors 714 collects the material that is being released from the crop processing components of combine 400.

Mitigators 702 can also include burier(s) 716 that are configured to bury the weed seeds to a threshold depth in the ground. For instance, a burier can be attached to, pulled by, or otherwise supported by combine 400. The burier follows the chaff ejection components which can either spread the chaff on the ground, drop the chaff in a windrow, or otherwise. The buriers operate to bury the ejected chaff, and thus the weed seeds, to a threshold depth in the ground (e.g., twelve inches or deeper) which inhibits germination of the weed seeds. Also, it is noted that a burier can comprise a separate machine that follows combine 400. Seed mitigators 702 can include other types of mitigators as well. This is represented by block 718.

System 570 is also illustrated as including communication/control logic 720 configured to communicate with other items in combine 400 and/or generate control signals to control weed seed mitigators 702. Of course, system 570 can include other items 722 as well.

FIGS. 8A and 8B (collectively referred to as FIG. 8) is a flow diagram 800 illustrating an example operation to generate weed seed locations and perform pre-emergent mitigation during a harvesting operation. For sake of illustration, but not by limitation, FIG. 8 will be described in the context of combine 400 illustrated in FIGS. 4 and 5.

At block 801, a prior generated weed map of the target field (if any) is obtained. For example, this can include receiving weed map 618 shown in FIG. 6. At block 802, a location and/or heading/path of combine 400 is obtained. For instance, this can be based on the sensor signals 620 which illustrate a current position and direction of combine 400.

At block 803, images of plants in the path of combine 400 (e.g., forthcoming plants that are expected to reach the crop processing functionality of the combine) are obtained. For instance, this can include mono images (block 804), stereo images (block 805), or other images (block 806). For example, the images obtained at block 803 can be received as images 626 from image capture system 124 that is on-board combine 400 and/or carried on a separate machine, such as UAV that operates ahead of combine 400 in the field.

At block 807, image classification is performed on the images obtained at block 803. Examples of image classification are discussed above. Briefly, however, the image classification performed at block 807 classifies areas of the image as representing weed plants, crop plants, etc. Based on this, a weed map is obtained at block 808. As represented at block 809, the weed map can include the generation of a new weed map and/or the modification of an existing map.

For example, the image classification performed at block **807** can be utilized to modify the weed map obtained at block **801**. The weed map can be obtained in other ways as well. This is represented at block **810**.

At block **811**, weed areas are identified based on the weed map obtained at block **808**. The identified weed areas can include a spatial boundary that identifies its relative position on the field, as well as weed density information that identifies the density of the weeds within the weed area.

At block **812**, each weed area is correlated to positions of the machine when the header **102** is within the weed area (i.e., when the header is actively cutting within the weed area).

For sake of illustration, FIG. **9** illustrates an example weed area **813** on a field **814** being harvested by combine **400**. The correlation at block **812** identifies the position of combine **400** when the header first reaches the beginning edge **815** of weed area **813** until the header passes the trailing edge **816**. A treatment zone is defined between edges **815** and **816** (and corresponding to the width of the header). Thus, for machine positions in which the header is within the treatment zone, the header is cutting the weed area, and is thus gathering weed seeds along with crop plants from that area of the field.

As also shown, mitigation system **570** follows combine **400** (e.g., is attached to the combine **400**, is pulled by combine **400**, is pulled/carried by a separate machine, etc.) and is configured to perform a weed seed mitigation operation. For example, system **570** can include a trailer or container that is configured to collect the chaff or residue ejected from the rear of combine **400**, bury the chaff, burn the chaff, chemically treat the chaff, etc. In one example, a bagger collects the chaff into bags that are deposited or dropped on the field and later collected. System **570** can tag (e.g., with markings, barcodes, radio-frequency identification (RFID) tags, the bags with information on the collected chaff (e.g., location, type of material, quantity, weed seed composition, etc.). This information can be read from the tags during subsequent processing.

Referring again to FIG. **8**, at block **817**, weed seed movement through the combine is determined. As discussed above, this, in one example, is based on machine delay compensation that represents the amount of time that it takes the weed seeds to move from a given point in the combine (e.g., the header cutting in the treatment zone) to a weed seed mitigator (e.g., collector, crusher, burner, etc.). This machine delay compensation is utilized by control system **540** to determine when to activate the corresponding mitigator to minimize operation of the mitigator. In other words, this operates to prevent or at least reduce operation of the mitigator during times when the mitigator is receiving material that does not include weed seeds or has weed seeds below a threshold. Thus, the mitigator is activated based on the spatial location of mitigator relative to the weed seed area. This can operate to reduce the associated costs of operating the mitigator. That is, some mitigators, such as seed crushers, have high operating costs in terms of power consumption, reduced efficiencies, wear and tear on the components, etc.

In one example, the weed movement within combine **400** is estimated, for example based on the machine operation model and/or current settings (e.g., settings of thresher **556**, cleaner **558**, chaff processing **560**, etc.). This is represented by block **818**. Alternatively, or in addition, weed seed movement can be detected based on signals received from on-board sensors. This is represented by block **819**. For example, on-board sensors such as electromagnetic sensors

(block **820**), capacitive sensors (block **821**), impact sensors (block **822**), or other types of sensors (block **823**) can be utilized.

In some examples, chaff material such as leaves, stalks, or other residue material are typically less dense than weed seeds and have lower concentrations of protein and/or oil. Leaves and stalks are typically composed of only carbohydrates, and possibly a small amount of chlorophyll and water. This information can be leveraged from the signals received from on-board sensors **819** to determine the presence of weed seeds in the material flow.

Referring to FIGS. **10A** and **10B** (collectively referred to as FIG. **10**), an example image **1000** is received from an electromagnetic sensor that uses electromagnetic radiation that is reflected, scattered, transmitted, or absorbed by components of the material flow. In one example, image **1000** includes a grayscale image of the electromagnetic transmission through a material flow **1002**.

Due to their differences in composition (e.g., as noted above, seeds typically have higher concentrations of oil and protein), chaff and seeds can have different electromagnetic responses that can be detected and utilized to identify weed seeds in material flow **1002**. Because seeds are denser than chaff, they show up darker in image **1000**. Similarly, size can be determined from the image and/or attenuation of the electromagnetic source signal (e.g., darker portions of the image can represent thicker material).

FIG. **10B** shows image **1000** after gray objects have been classified as soybean seed (identified by reference numeral **1006**). Similarly, palmer amaranth is identified at reference numeral **1008** and common ragweed seed is identified at reference numeral **1010** using shape and size information. In one example, texture as well as non-visible spectra is utilized to determine density and/or reflectance difference. Alternatively, or in addition, active (modulated/pulsed) illumination is reposed in different spectral bands. In one example of FIG. **10B**, chaff material represented at reference numeral **1012** is ignored.

It is noted that any type of classifier can be utilized for classifying the material in material flow **1002**. For example, a rules-based classifier can classify weed seeds based on size, shape, and/or color. In another example, a neural network can be trained to identify seeds.

Referring again to FIG. **8**, at block **826** weed seed mapping system **606** determines a concentration of weed seed within the plant material. That is, one example, block **826** determines the ratio of crops to weeds within the harvested material. At block **827**, control logic **610** can control user interface mechanism **642** to output an indication of the weed seed processing to operator **130**. For instance, an augmented view of the weed seed can be provided.

At block **828**, one or more weed seed mitigators are controlled by control logic **610** generating control signals **638** to weed mitigation system **570**. As noted above, this can include any of a wide variety of operations performed by combine **400**. For example, the chaff treatment system can be controlled at block **829**. Alternatively, or in addition, a seed crusher is controlled at block **830**, a seed burner is controlled at block **831**, a seed collector/bagger is controlled at block **832**, a seed burier is controlled at block **833**, and/or chaff discharge can be controlled at block **834**.

For example, the chaff discharge **834** can be switched from a spreading mode to a windrowing mode. For instance, when high concentrations of weed seeds are about to be discharged from combine **400**, the discharge settings can be

set to place the material in a windrow for subsequent burning, burying, and/or pickup (by combine **400** or another machine).

Of course, other types of weed seed mitigators can be controlled as well. This is represented by block **835**.

In one example, the control at block **828** includes adjusting header **402** of combine **400**. This is represented at block **836**. For example, in response to detecting the presence of weed plants in the path of the header, header **402** can be lowered to ensure that the crops are cut and processed through combine **400**, so their seeds do not remain on the field. Alternatively, header **402** can be raised so that the weed area is not harvested. The weed area can be treated on-site (e.g., mechanically) to minimize weed seed spread.

In another example, the thresher settings can be adjusted at block **837**. Alternatively, or in addition, the machine speed can be adjusted at block **838**. For example, it may be that seed crusher **830** performs better at lower machine speeds. Thus, the machine can be controlled to a desired speed during operation of the weed seed mitigator.

Also, it is noted that the control at block **828** can be automatic, for example based on weed seed concentration thresholds. This is represented at block **839**. For example, the weed seed concentration determined at block **826** can be compared against one or more thresholds and, if a particular threshold is met, one of weed seed mitigators **702** can be activated. Alternatively, or in addition, the control at block **828** can be manual, for example based on operator input. This is represented by block **840**.

At block **842**, the weed seed locations in the field are generated. For example, the weed seed locations can be generated based on one or more factors that affect the movement of the weed seeds once they are released from combine **400**. For instance, this can be based on the machine speed, location and/or orientation when released. This is represented by block **844**. Alternatively, or in addition, the locations of the weed seed can be determined based on environmental factors, such as air density, wind, rain, surface water, surface topology, etc. This is represented at block **846**. Also, the weed seed locations can be generated based on chaff spreader data (e.g., data **634**). This is represented at block **847**.

The weed seed locations generated at block **842** can be stored on-board combine **400** (block **848**) in data store **534**. Alternatively, or in addition, the weed seed locations can be sent to a remote system, such as system **114**, other agricultural machines **117**, and/or another machine/system (block **849**). Of course, the weed seed locations can be generated and utilized in other ways as well. This is represented by block **850**.

At block **852**, other machines are controlled to perform pre-emergent field operations based on the weed seed locations. For instance, a tilling machine can be controlled to perform a tilling operation in treatment areas of the field that correspond to the weed seed locations. This is represented by block **853**. In another example, a chemical application machine (e.g., a sprayer) is controlled to chemically treat the weed seed locations to destroy the seeds, stimulate early germination, etc. This is represented by block **854**. In another example, at block **855**, predation can be enhanced by an application of a substance to the weed seeds. In another example, selective planting can be performed in the next growing season. This is represented by block **856**. For instance, different crop planting prescriptions can be utilized based on the identification of weed seed areas in the field and/or their corresponding weed seed densities.

In another example, operation of a grain processor can be controlled based on the crop/weed ratio. This is represented by block **857**. For example, when a grain cart is filled by combine **400**, the weed seed concentration information determined at block **826** can be utilized to determine a subsequent grain processing operation to remove the weed seeds.

Of course, pre-emergent operations can be performed in other ways as well. This is represented by block **858**.

As noted above, in some examples weed plants include a second crop intercropped with a first crop and undergoing simultaneous harvest. In accordance with one example of FIG. **8**, the weed areas identified at block **811** (e.g., based on the prior-generated weed map obtained at block **801**, the weed map obtained at block **808**, etc.) identify areas of the field that contain the second crop. For instance, the weed seed locations **640** (shown in FIG. **6**) represent second crop locations and can include as-planted maps, crop maps from aerial images, etc.

Further, the operations controlled at blocks **828** and/or **852** can be configured to nondestructively collect and segregate the second crop ("the weed") from the first crop. For sake of illustration, at block **832** a seed collector/bagger is configured to segregate and collect the second crop. In another example, adjustments can be made (e.g., blocks **836**, **837**, etc.) to collect and keep both crops together for later separation with minimal (or at least reduced) harvest losses in the field.

In one example, the seed collector/bagger is equipped with a crop sensor for measuring the yield or other attributes (e.g., moisture, oil, protein, starch, etc.) of the second crop. These yield and crop attribute measurements can be displayed to an operator (in place of or in addition to the those of the first crop), georeferenced and stored on a map, and/or used for machine control.

It can thus be seen that the present system provides a number of advantages. For example, but not by limitation, the present system provides site-specific agricultural operations to mitigate weeds by identifying, before emergence, the locations of the weed seeds. Further, the weed seed locations can be accurately determined by taking into account machine data, environmental data, and/or any other data utilized to model weed seed movement. Using those weed seed locations, the weeds (including herbicide resistant varieties) can be mitigated. This can increase yields, while reducing the application of chemicals to the fields and/or machine operations (e.g., sprayer operation over the field to chemically treat weeds post-emergence. This, in turn, can decrease power/fuel consumption, reduced machine wear and tear, and/or otherwise increase efficiencies/productivity of the agricultural operations.

It will be noted that the above discussion has described a variety of different systems, components and/or logic. It will be appreciated that such systems, components and/or logic can be comprised of hardware items (such as processors and associated memory, or other processing components, some of which are described below) that perform the functions associated with those systems, components and/or logic. In addition, the systems, components and/or logic can be comprised of software that is loaded into a memory and is subsequently executed by a processor or server, or other computing component, as described below. The systems, components and/or logic can also be comprised of different combinations of hardware, software, firmware, etc., some examples of which are described below. These are only some examples of different structures that can be used to

form the systems, components and/or logic described above. Other structures can be used as well.

The present discussion has mentioned processors, processing systems, controllers and/or servers. In one example, these can include computer processors with associated memory and timing circuitry, not separately shown. They are functional parts of the systems or devices to which they belong and are activated by, and facilitate the functionality of the other components or items in those systems.

Also, a number of user interface displays have been discussed. They can take a wide variety of different forms and can have a wide variety of different user actuatable input mechanisms disposed thereon. For instance, the user actuatable input mechanisms can be text boxes, check boxes, icons, links, drop-down menus, search boxes, etc. They can also be actuated in a wide variety of different ways. For instance, they can be actuated using a point and click device (such as a track ball or mouse). They can be actuated using hardware buttons, switches, a joystick or keyboard, thumb switches or thumb pads, etc. They can also be actuated using a virtual keyboard or other virtual actuators. In addition, where the screen on which they are displayed is a touch sensitive screen, they can be actuated using touch gestures. Also, where the device that displays them has speech recognition components, they can be actuated using speech commands.

A number of data stores have also been discussed. It will be noted they can each be broken into multiple data stores. All can be local to the systems accessing them, all can be remote, or some can be local while others are remote. All of these configurations are contemplated herein.

Also, the figures show a number of blocks with functionality ascribed to each block. It will be noted that fewer blocks can be used so the functionality is performed by fewer components. Also, more blocks can be used with the functionality distributed among more components.

FIG. 11 is a block diagram of one example of the architecture shown in FIG. 5, where machine 400 communicates with elements in a remote server architecture 900. In an example, remote server architecture 900 can provide computation, software, data access, and storage services that do not require end-user knowledge of the physical location or configuration of the system that delivers the services. In various examples, remote servers can deliver the services over a wide area network, such as the internet, using appropriate protocols. For instance, remote servers can deliver applications over a wide area network and they can be accessed through a web browser or any other computing component. Software or components shown in FIG. 5 as well as the corresponding data, can be stored on servers at a remote location. The computing resources in a remote server environment can be consolidated at a remote data center location or they can be dispersed. Remote server infrastructures can deliver services through shared data centers, even though they appear as a single point of access for the user. Thus, the components and functions described herein can be provided from a remote server at a remote location using a remote server architecture. Alternatively, they can be provided from a conventional server, or they can be installed on client devices directly, or in other ways.

In the example shown in FIG. 11, some items are similar to those shown in FIG. 5 and they are similarly numbered. FIG. 11 specifically shows that system 124, system 126, system 540, and/or data store 534 can be located at a remote server location 902. Therefore, agricultural machine 400, machine(s) 115, machine(s) 117, and/or system(s) 114 access those systems through remote server location 902.

FIG. 11 also depicts another example of a remote server architecture. FIG. 11 shows that it is also contemplated that some elements of FIG. 5 are disposed at remote server location 902 while others are not. By way of example, data store 534 can be disposed at a location separate from location 902, and accessed through the remote server at location 902. Alternatively, or in addition, one or more of systems 124, 126, and 540 can be disposed at location(s) separate from location 902, and accessed through the remote server at location 902.

Regardless of where they are located, they can be accessed directly by agricultural machine 400, through a network (either a wide area network or a local area network), they can be hosted at a remote site by a service, or they can be provided as a service, or accessed by a connection service that resides in a remote location. Also, the data can be stored in substantially any location and intermittently accessed by, or forwarded to, interested parties. For instance, physical carriers can be used instead of, or in addition to, electromagnetic wave carriers. In such an example, where cell coverage is poor or nonexistent, another mobile machine (such as a fuel truck) can have an automated information collection system. As the agricultural machine comes close to the fuel truck for fueling, the system automatically collects the information from the machine or transfers information to the machine using any type of ad-hoc wireless connection. The collected information can then be forwarded to the main network as the fuel truck reaches a location where there is cellular coverage (or other wireless coverage). For instance, the fuel truck may enter a covered location when traveling to fuel other machines or when at a main fuel storage location. All of these architectures are contemplated herein. Further, the information can be stored on the agricultural machine until the agricultural machine enters a covered location. The agricultural machine, itself, can then send and receive the information to/from the main network.

It will also be noted that the elements of FIG. 5, or portions of them, can be disposed on a wide variety of different devices. Some of those devices include servers, desktop computers, laptop computers, tablet computers, or other mobile devices, such as palm top computers, cell phones, smart phones, multimedia players, personal digital assistants, etc.

FIG. 12 is a simplified block diagram of one illustrative example of a handheld or mobile computing device that can be used as a user's or client's handheld device 16, in which the present system (or parts of it) can be deployed. For instance, a mobile device can be deployed in the operator compartment of agricultural machine 400 or as remote computing system 114. FIGS. 13-14 are examples of handheld or mobile devices.

FIG. 12 provides a general block diagram of the components of a client device 16 that can run some components shown in FIG. 5, that interacts with them, or both. In the device 16, a communications link 13 is provided that allows the handheld device to communicate with other computing devices and under some embodiments provides a channel for receiving information automatically, such as by scanning. Examples of communications link 13 include allowing communication through one or more communication protocols, such as wireless services used to provide cellular access to a network, as well as protocols that provide local wireless connections to networks.

In other examples, applications can be received on a removable Secure Digital (SD) card that is connected to an interface 15. Interface 15 and communication links 13

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communicate with a processor 17 (which can also embody processors or servers from previous FIGS.) along a bus 19 that is also connected to memory 21 and input/output (I/O) components 23, as well as clock 25 and location system 27.

I/O components 23, in one example, are provided to facilitate input and output operations. I/O components 23 for various embodiments of the device 16 can include input components such as buttons, touch sensors, optical sensors, microphones, touch screens, proximity sensors, accelerometers, orientation sensors and output components such as a display device, a speaker, and or a printer port. Other I/O components 23 can be used as well.

Clock 25 illustratively comprises a real time clock component that outputs a time and date. It can also, illustratively, provide timing functions for processor 17.

Location system 27 illustratively includes a component that outputs a current geographical location of device 16. This can include, for instance, a global positioning system (GPS) receiver, a LORAN system, a dead reckoning system, a cellular triangulation system, or other positioning system. It can also include, for example, mapping software or navigation software that generates desired maps, navigation routes and other geographic functions.

Memory 21 stores operating system 29, network settings 31, applications 33, application configuration settings 35, data store 37, communication drivers 39, and communication configuration settings 41. Memory 21 can include all types of tangible volatile and non-volatile computer-readable memory devices. It can also include computer storage media (described below). Memory 21 stores computer readable instructions that, when executed by processor 17, cause the processor to perform computer-implemented steps or functions according to the instructions. Processor 17 can be activated by other components to facilitate their functionality as well.

FIG. 13 shows one example in which device 16 is a tablet computer 950. In FIG. 13, computer 950 is shown with user interface display screen 952. Screen 952 can be a touch screen or a pen-enabled interface that receives inputs from a pen or stylus. It can also use an on-screen virtual keyboard. Of course, it might also be attached to a keyboard or other user input device through a suitable attachment mechanism, such as a wireless link or USB port, for instance. Computer 950 can also illustratively receive voice inputs as well.

FIG. 14 shows that the device can be a smart phone 71. Smart phone 71 has a touch sensitive display 73 that displays icons or tiles or other user input mechanisms 75. Mechanisms 75 can be used by a user to run applications, make calls, perform data transfer operations, etc. In general, smart phone 71 is built on a mobile operating system and offers more advanced computing capability and connectivity than a feature phone.

Note that other forms of the devices 16 are possible.

FIG. 15 is one example of a computing environment in which elements of FIG. 5, or parts of it, (for example) can be deployed. With reference to FIG. 15, an example system for implementing some embodiments includes a computing device in the form of a computer 1010. Components of computer 1010 may include, but are not limited to, a processing unit 1020 (which can comprise processors or servers from previous FIGS.), a system memory 1030, and a system bus 1021 that couples various system components including the system memory to the processing unit 1020. The system bus 1021 may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus

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architectures. Memory and programs described with respect to FIG. 5 can be deployed in corresponding portions of FIG. 15.

Computer 1010 typically includes a variety of computer readable media. Computer readable media can be any available media that can be accessed by computer 1010 and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media. Computer storage media is different from, and does not include, a modulated data signal or carrier wave. It includes hardware storage media including both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computer 1010. Communication media may embody computer readable instructions, data structures, program modules or other data in a transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

The system memory 1030 includes computer storage media in the form of volatile and/or nonvolatile memory such as read only memory (ROM) 1031 and random access memory (RAM) 1032. A basic input/output system 1033 (BIOS), containing the basic routines that help to transfer information between elements within computer 1010, such as during start-up, is typically stored in ROM 1031. RAM 1032 typically contains data and/or program modules that are immediately accessible to and/or presently being operated on by processing unit 1020. By way of example, and not limitation, FIG. 15 illustrates operating system 1034, application programs 1035, other program modules 1036, and program data 1037.

The computer 1010 may also include other removable/non-removable volatile/nonvolatile computer storage media. By way of example only, FIG. 15 illustrates a hard disk drive 1041 that reads from or writes to non-removable, nonvolatile magnetic media, an optical disk drive 1055, and non-volatile optical disk 1056. The hard disk drive 1041 is typically connected to the system bus 1021 through a non-removable memory interface such as interface 1040, and optical disk drive 1055 is typically connected to the system bus 1021 by a removable memory interface, such as interface 1050.

Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (e.g., ASICs), Application-specific Standard Products (e.g., ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

The drives and their associated computer storage media discussed above and illustrated in FIG. 15, provide storage of computer readable instructions, data structures, program modules and other data for the computer 1010. In FIG. 15,

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for example, hard disk drive **1041** is illustrated as storing operating system **1044**, application programs **1045**, other program modules **1046**, and program data **1047**. Note that these components can either be the same as or different from operating system **1034**, application programs **1035**, other program modules **1036**, and program data **1037**.

A user may enter commands and information into the computer **1010** through input devices such as a keyboard **1062**, a microphone **1063**, and a pointing device **1061**, such as a mouse, trackball or touch pad. Other input devices (not shown) may include a joystick, game pad, satellite dish, scanner, or the like. These and other input devices are often connected to the processing unit **1020** through a user input interface **1060** that is coupled to the system bus, but may be connected by other interface and bus structures. A visual display **1091** or other type of display device is also connected to the system bus **1021** via an interface, such as a video interface **1090**. In addition to the monitor, computers may also include other peripheral output devices such as speakers **1097** and printer **1096**, which may be connected through an output peripheral interface **1095**.

The computer **1010** is operated in a networked environment using logical connections (such as a local area network—LAN, or wide area network—WAN or a controller area network—CAN) to one or more remote computers, such as a remote computer **1080**.

When used in a LAN networking environment, the computer **1010** is connected to the LAN **1071** through a network interface or adapter **1070**. When used in a WAN networking environment, the computer **1010** typically includes a modem **1072** or other means for establishing communications over the WAN **1073**, such as the Internet. In a networked environment, program modules may be stored in a remote memory storage device. FIG. **15** illustrates, for example, that remote application programs **1085** can reside on remote computer **1080**.

It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein.

Example 1 is an agricultural harvesting machine comprising:

- a crop processing functionality configured to engage crop in a field, perform a crop processing operation on the crop, and move the processed crop to a harvested crop repository; and
- a control system configured to:
 - identify a weed seed area indicating presence of weed seeds; and
 - generate a control signal associated with a pre-emergence weed seed treatment operation based on the identified weed seed area.

Example 2 is the agricultural harvesting machine of any or all previous examples, wherein the control signal controls a pre-emergence weed seed mitigator to perform the pre-emergence weed seed treatment operation.

Example 3 is the agricultural harvesting machine of any or all previous examples, wherein the pre-emergence weed seed mitigator is on-board the agricultural harvesting machine.

Example 4 is the agricultural harvesting machine of any or all previous examples, wherein the pre-emergence weed seed mitigator is separate from the agricultural harvesting machine.

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Example 5 is the agricultural harvesting machine of any or all previous examples, wherein the pre-emergence weed seed mitigator comprises a weed seed collector configured to collect the weed seeds.

Example 6 is the agricultural harvesting machine of any or all previous examples, wherein the pre-emergence weed seed treatment operation devitalizes the weed seeds.

Example 7 is the agricultural harvesting machine of any or all previous examples, wherein the pre-emergence weed seed mitigator comprises at least one of:

- a weed seed burier configured to bury the weed seeds in the field,
- a weed seed crusher configured to mechanically crush the weed seeds,
- a thermal weed seed treatment device configured to thermally treat the weed seeds, or
- a chemical weed seed treatment device configured to chemically treat the weed seeds.

Example 8 is the agricultural harvesting machine of any or all previous examples, wherein the control signal controls at least one of:

- a display device to display an indication of the weed seed area to an operator, or
- a data storage device to store an indication of the identified weed seed area.

Example 9 is the agricultural harvesting machine of any or all previous examples, wherein the weed seed area is identified based on an a priori weed map.

Example 10 is the agricultural harvesting machine of any or all previous examples, and further comprising:

- an imaging sensor, wherein the weed seed area is identified based on one or more images, obtained from the imaging sensor, of the field in a path of the agricultural harvesting machine.

Example 11 is the agricultural harvesting machine of any or all previous examples, and further comprising:

- a weed seed detector configured to detect weed seeds, wherein the weed seed area is identified based on a weed seed presence signal received from the weed seed detector.

Example 12 is the agricultural harvesting machine of any or all previous examples, wherein the weed seed area is identified based on a weed seed movement model that models movement of the weed seeds.

Example 13 is the agricultural harvesting machine of any or all previous examples, wherein the weed seed movement model is based on machine delay compensation that represents movement of the weed seeds through the agricultural harvesting machine.

Example 14 is the agricultural harvesting machine of any or all previous examples, wherein the weed seed area is identified based on chaff spreader data received from a chaff spreader sensor on the agricultural harvesting machine.

Example 15 is the agricultural harvesting machine of any or all previous examples, wherein the weed seed area is identified based on at least one of:

- environment data representing an environment of the field,
- terrain data representing a terrain of the field, or
- machine data representing machine operating characteristics.

Example 16 is a method performed by an agricultural machine, the method comprising:

- obtaining a weed seed movement model that models movement of weed seeds during a crop processing operation that engages crop in a field and moves the processed crop to a harvested crop repository;

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identifying a weed seed area based on the weed seed movement model; and
generating a control signal associated with a pre-emergence weed seed treatment operation based on the identified weed seed area.

Example 17 is the method of any or all previous examples, and further comprising:

obtaining a weed map that identifies weed area in the field; and

applying the weed seed movement model to determining movement of the weed seeds from the identified weed area; and

identifying the weed area based on the determined movement.

Example 18 is the method of any or all previous examples, wherein the weed seed movement model is based on at least one of:

environment data representing an environment of the field,

terrain data representing a terrain of the field, or
machine data representing machine operating characteristics.

Example 19 is a computing system comprising:

at least one processor; and

memory storing instructions executable by the at least one processor, wherein the instructions, when executed, cause the computing system to:

identify a weed seed area indicating presence of weed seeds on a field during a harvesting operation that harvests crops from the field; and

generate a control signal associated with a pre-emergence weed seed treatment operation based on the identified weed seed area.

Example 20 is the computing system of any or all previous examples, wherein the instructions cause the computing system to identify the weed seed area based on a weed seed movement model models the movement of weed seeds based on at least one of:

environment data representing an environment of the field,

terrain data representing a terrain of the field, or
machine data representing machine operating characteristics.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

What is claimed is:

1. An agricultural harvesting machine comprising:

crop processing functionality configured to engage crop in a field, perform a crop processing operation on the crop to obtain processed crop, and move the processed crop to a harvested crop repository; and

a control system configured to:

identify a weed seed area indicating presence of weed seeds based on a weed seed movement model that projects likely locations of the weed seeds given a location of one or more weed plants in the field that produced the weed seeds, wherein the weed seed movement model represents machine delay compensation and models a distance, relative to a surface of the field, between:

when a weed plant that carried the weed seeds is cut by the agricultural harvesting machine, and

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the weed seeds are discharged on to the surface of the field by the agricultural harvesting machine; and
generate a control signal associated with a pre-emergence weed seed treatment operation on the weed seeds, prior to emergence of the weed seeds, based on the weed seed area.

2. The agricultural harvesting machine of claim 1, wherein the weed seed area is identified based on one or more of:

environment data representing an environment of the field, or

terrain data representing a terrain of the field.

3. The agricultural harvesting machine of claim 1, wherein the weed seed area is identified based on machine data representing machine operating characteristics.

4. The agricultural harvesting machine of claim 1, wherein the control signal controls a pre-emergence weed seed mitigator to perform the pre-emergence weed seed treatment operation on the weed seeds prior to emergence of the weed seeds.

5. The agricultural harvesting machine of claim 4, wherein the pre-emergence weed seed mitigator is on-board the agricultural harvesting machine.

6. The agricultural harvesting machine of claim 4, wherein the pre-emergence weed seed mitigator comprises a mitigator machine that is separate from the agricultural harvesting machine, wherein the control system is configured to transmit the control signal to the mitigator machine.

7. The agricultural harvesting machine of claim 4, wherein the pre-emergence weed seed mitigator comprises: a weed seed collector configured to collect the weed seeds.

8. The agricultural harvesting machine of claim 4, wherein the pre-emergence weed seed treatment operation devitalizes the weed seeds.

9. The agricultural harvesting machine of claim 8, wherein the pre-emergence weed seed mitigator comprises at least one of:

a weed seed burier configured to bury the weed seeds in the field,

a weed seed crusher configured to mechanically crush the weed seeds,

a thermal weed seed treatment device configured to thermally treat the weed seeds, or

a chemical weed seed treatment device configured to chemically treat the weed seeds.

10. The agricultural harvesting machine of claim 1, wherein the control signal controls at least one of:

a display device to display an indication of the weed seed area to an operator, or

a data storage device to store an indication of the weed seed area.

11. The agricultural harvesting machine of claim 1, wherein the weed seed area is identified based on an a priori weed map.

12. The agricultural harvesting machine of claim 1, and further comprising:

an imaging sensor, wherein the weed seed area is identified based on one or more images, obtained from the imaging sensor, of the field in a path of the agricultural harvesting machine.

13. The agricultural harvesting machine of claim 1, and further comprising:

a weed seed detector configured to detect weed seeds, wherein the weed seed area is identified based on a weed seed presence signal received from the weed seed detector.

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14. A method performed by an agricultural harvesting machine, the method comprising:

operating crop processing functionality of the agricultural harvesting machine to engage crop in a field, perform a crop processing operation on the crop to obtain processed crop, and move the processed crop to a harvested crop repository;

identifying a weed seed area indicating presence of weed seeds based on a weed seed movement model that projects likely locations of the weed seeds given a location of one or more weed plants in the field that produced the weed seeds, wherein the weed seed movement model represents machine delay compensation and models a distance, relative to a surface of the field, between:

when a weed plant that carried the weed seeds is cut by the agricultural harvesting machine, and

the weed seeds are discharged on to the surface of the field by the agricultural harvesting machine; and

generating a control signal associated with a pre-emergence weed seed treatment operation based on the weed seed area.

15. The method of claim 14, and further comprising:

obtaining a weed map that identifies a weed area in the field;

applying the weed seed movement model to determining a movement of the weed seeds from the weed area; and identifying the weed seed area based on the movement of the weed seeds.

16. The method of claim 14, wherein

the weed seed movement model is based on at least one of:

environment data representing an environment of the field during the crop processing operation, or terrain data representing a terrain of the field.

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17. An agricultural harvesting machine comprising:

crop processing functionality configured to engage crop in a field, perform a crop processing operation on the crop to obtain processed crop, and move the processed crop to a harvested crop repository; and

a control system configured to:

identify a weed seed area indicating presence of weed seeds based on a weed seed movement model that projects likely locations of the weed seeds given a location of one or more weed plants in the field that produced the weed seeds, wherein the weed seed area is identified based on chaff spreader data received from a chaff spreader sensor on the agricultural harvesting machine, wherein the chaff spreader data represents chaff ejected from the agricultural harvesting machine during the crop processing operation; and

generate a control signal associated with a pre-emergence weed seed treatment operation on the weed seeds, prior to emergence of the weed seeds, based on the weed seed area.

18. The agricultural harvesting machine of claim 17, wherein the control signal controls a pre-emergence weed seed mitigator to perform the pre-emergence weed seed treatment operation on the weed seeds prior to emergence of the weed seeds.

19. The agricultural harvesting machine of claim 18, wherein the pre-emergence weed seed mitigator is on-board the agricultural harvesting machine.

20. The agricultural harvesting machine of claim 17, wherein the control signal controls at least one of:

a display device to display an indication of the weed seed area, or

a data storage device to store an indication of the weed seed area.

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